Topics on Fundamental Science

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Lecture Book

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Part I

Essential Calculus

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Physics is written in this grand book . . . which stands continually open to our gaze, but it cannot be understood unless one first learns to comprehend the language and interpret the characters in which it is written. It is written in the language of mathematics, and its characters are triangles, circles, and other geometrical figures, without which it is humanly impossible to understand a single word of it; without these, one is wandering around in a dark labyrinth.

(Galilei, Galileo: Il Saggiatore, Chapter 6)



Part II

Ordinary Differential Equations

Part Contents

. . . If x, instead of being a function of two variables x and y, should contain only x, this would be no more than an ordinary differential equation, which, being integrated, would give . . .

(Jean-Louis Boucharlat and Ralph Blakelock, An Elementary Treatise on the Differential and Integral Calculus where the term "Ordinary Differential Equations" is first used.)



Part III

Linear Algebra & Vector Calculus

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Mathematics is the art of reducing any problem to linear algebra.

(William Stein - Mathematician, the lead developer of SageMath.)

Chapter 1

Linear Algebra I - Fundamentals

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Linear Algebra is an extensive subject covering wide topics such as:

> *vectors*, *matrices*, *determinants*, *systems of linear equations*, *vector spaces*, *linear transformations*, *eigenvalue problems*,

and much more.[1] As an area of study it has a broad appeal in that it has many applications in engineering, physics, geometry, computer science, economics, and other areas.[2]

[1] As there are numerous textbooks and publications dedicated to this topic.

[2] It also contributes to a deeper understanding of mathematics itself.

Matrices, which are **rectangular arrays of numbers or functions**, and vectors are the main tools of linear algebra. Matrices are important as they allow us express large amounts of data and functions in an organized and concise form. In addition, as matrices are single objects, we denote them by **single letters** and calculate with them directly.

All these features have made matrices and vectors very popular for expressing scientific and mathematical ideas.

1.1 Matrices, Vectors: Addition and Scalar Multiplication

As an organizational note, the fundamental concepts and rules of matrix and vector algebra are introduced in Section 1.1 and are then followed by **linear systems**,[3] a main application, as explained in detail in Section 1.2.

[3] also called systems of linear equations.

As a gentle beginning to this topic, let us start by taking a leisurely look at matrices before we formalize our discussion. A matrix is a rectangular array of numbers or functions which we will enclose in brackets.

For example,

$$\begin{bmatrix} 0.5 & 2 & -7 \\ 0 & -0.8 & 26 \end{bmatrix}, \quad \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \quad \begin{bmatrix} e^{-2x} & 2x^4 \\ e^{8x} & 23x \end{bmatrix}, \quad \begin{bmatrix} a_1 & a_2 & a_3 \end{bmatrix}, \quad \text{and} \quad \begin{bmatrix} 4 \\ \frac{1}{5} \end{bmatrix} \tag{1.1}$$

are all matrices. The numbers, or functions, are called entries or, less commonly, elements of the matrix. The first matrix in Eq. (1.1) has two (2) **rows**, which are the horizontal lines of entries.

Furthermore, it has three (3) columns, which are the vertical lines of entries. The second and third matrices are square matrices, which means that each has as many rows as columns 3 and 2, respectively. The entries of the 2nd matrix have two (2) indices, signifying their location within the matrix.

i. The first index is the number of the row, and

ii. the second is the number of the column, so that together the entry's position is uniquely identified.

   For example, a_{23}[4] is in **Row 2** and **Column 3**, etc.

[4] This is read as "a two three".

> The notation is standard and applies to all matrices, including those which are **NOT** square.

Matrices having only a single row or column are called **vectors**. Therefore, the 4th matrix in Eq. (1.1) has just one row and is called a row vector. The last matrix in Eq. (1.1) has just one column and is called a **column vector**. Because the goal of the indexing of entries was to uniquely identify the position of an element within a matrix, one index suffices for vectors, whether they are row or column vectors. Thus, the third entry of the row vector in Eq. (1.1) is denoted by a_3.

Matrices are handy for storing and processing data in applications. Consider the following two common examples.

Linear Systems, a Major Application of Matrices 1.1 Example

We are given a system of linear equations, briefly a linear system, such as

$$\begin{aligned} 4x_1 + 6x_2 + 9x_3 &= 6 \\ 6x_1 - 2x_3 &= 20 \\ 5x_1 - 8x_2 + x_3 &= 10 \end{aligned}$$

where x_1, x_2, x_3 are the **unknowns**. We form the **coefficient matrix**, call it $\mathbf{A}$, by listing the coefficients of the unknowns in the position in which they appear in the linear equations. In the second equation, there is no unknown x_2, which means that the coefficient of x_2 is 0 and hence in matrix $\mathbf{A}$, $a_{22} = 0$, Thus,

$$\mathbf{A} = \begin{bmatrix} 4 & 6 & 9 \\ 6 & 0 & -2 \\ 5 & -8 & 1 \end{bmatrix}. \qquad \text{We form another matrix} \qquad \tilde{\mathbf{A}} = \begin{bmatrix} 4 & 6 & 9 & 6 \\ 6 & 0 & -2 & 20 \\ 5 & -8 & 1 & 10 \end{bmatrix}$$

by augmenting $\mathbf{A}$ with the right sides of the linear system and call it the **augmented matrix** of the system. Since we can go back and recapture the system of linear equations directly from the augmented matrix $\tilde{\mathbf{A}}$, $\tilde{\mathbf{A}}$ contains all the information of the system and can therefore be used to solve the linear system.

This means that we can just use the augmented matrix to do the calculations needed to solve the system.[5] Meanwhile we can verify using substitution, that the solution is $x_1 = 3$, $x_2 = \frac{1}{2}$, $x_3 = -1$.

[5] This will play an important role in Section 1.2.

The notation x_1, x_2, x_3 for the unknowns is practical but **NOT** essential as we could have choose x, y, z or some other letters.

Sales Figures in Matrix Form 1.2 Example

As a useful application we can write sales figures for three (3) products; namely I, II, III, in a store on Monday (Mon), Tuesday (Tues), as a matrix:

$$\mathbf{A} = \begin{array}{c} \begin{matrix} \text{Mon} & \text{Tues} & \text{Wed} & \text{Thur} & \text{Fri} & \text{Sat} & \text{Sun} \end{matrix} \\ \begin{bmatrix} 40 & 33 & 81 & 0 & 21 & 47 & 33 \\ 0 & 12 & 778 & 50 & 50 & 96 & 90 \\ 10 & 0 & 0 & 27 & 43 & 78 & 56 \end{bmatrix} \end{array} \begin{matrix} \text{I} \\ \text{II} \\ \text{III} \end{matrix}$$

If the company has 10 stores, we can set up 10 such matrices, one for each store. Then, by adding corresponding entries of these matrices, we can get a matrix showing the total sales of each product on each day ■

1.1.1 General Concepts and Notations

Now it is time to formalize what we just have discussed.

i. We will denote matrices by **capital boldface letters** $\mathbf{A}, \mathbf{B}, \mathbf{C}, \cdots$, or by writing the general entry in brackets, i.e., $\mathbf{A} = [a_{ik}]$, and so on.

ii. By an $m \times n$ matrix[6] we mean a matrix with m rows and n columns and remember that **rows always come first**!

[6] This is read as m by n matrix.

> $m \times n$ is called the **size** of the matrix.

Therefore an $m \times n$ matrix is of the form

$$\mathbf{A} = [a_{jk}] = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{vmatrix} \tag{1.2}$$

The matrices in Eq. (1.1) are of sizes 2×3, 3×3, 2×2, 1×3, and 2×1, respectively.

Each entry in Eq. (1.2) has two (2) subscripts.

The first is the row number and the second is the column number.

Thus a_{21} is the entry in Row 2 and Column 1. If $m = n$, we call $\mathbf{A}$ an $n \times n$ square matrix. Then its diagonal containing the entries $a_{11}, a_{22}, \ldots, a_{nn}$ is called the **main diagonal** of $\mathbf{A}$. Thus the main diagonals of the two square matrices in Eq. (1.1) are a_{11}, a_{22}, a_{33} and e^{-x}, $4x$, respectively.

Square matrices are particularly important, as we shall see. A matrix of any size $m \times n$ is called a rectangular matrix; this includes square matrices as a special case.

1.1.2 Vectors

A **vector** is a matrix with only one row or column, where its entries are called the **components** of the vector. We will denote **vectors** by lowercase letters, $\mathbf{a}, \mathbf{b}, \cdots$ or by its general component in brackets, $\mathbf{a} = [a_i]$, and so on.[7] Our special vectors in Eq. (1.1) suggest that, a general, row vector is of the form

$$\mathbf{a} = \begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix}. \qquad \text{For instance,} \qquad \mathbf{a} = \begin{bmatrix} -3 & 6 & 0.2 & 0 & 1 \end{bmatrix}.$$

[7] It is at this point worth mentioning, different disciplines use different notation to symbolise a **vector**. The most prominent one is between mathematics, where they use a boldface lowercase character, and introductionary physics/engineer materials uses a normal font lower-case letter with arrow on top of it.

A **column vector**, on the other hand, is of the form

$$\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ \vdots \end{bmatrix}. \quad \text{For example,} \quad \mathbf{b} = \begin{bmatrix} 2 \\ 1 \\ -9 \end{bmatrix}.$$

1.1.3 Addition and Scalar Multiplication of Matrices and Vectors

What makes matrices and vectors really useful and particularly suitable for computers is the fact that **we can calculate with them almost as easily as with numbers**.

We now introduce rules for addition and for scalar multiplication[8] which were suggested by practical applications.

[8] i.e, multiplication by numbers

We first need the concept of equality.

Equality of Matrices — 1.1 Definition

Two matrices, namely $\mathbf{A} = [a_{ik}]$ and $\mathbf{B} = [b_{ik}]$ are said to be **equal**, written $\mathbf{A} = \mathbf{B}$, if and only if they have the **same size** and the **corresponding entries are equal**, that is, $a_{11} = b_{11}$, $a_{12} = b_{12}$, and so on.

Matrices **NOT** equal are called **different**, meaning different sized matrices are always different.

Equality of Matrices — 1.3 Example

Let

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 4 & 0 \\ 3 & -1 \end{bmatrix}.$$

Then

$$\mathbf{A} = \mathbf{B} \quad \text{if and only if} \quad \begin{aligned} a_{11} &= 4, & a_{12} &= 0, \\ a_{21} &= 3, & a_{22} &= -1. \end{aligned}$$

Addition of Matrices — 1.2 Definition

The **sum** of two (2) matrices, namely, $\mathbf{A} = [a_{jk}]$ and $\mathbf{B} = [b_{jk}]$ of the same size is written as $\mathbf{A} + \mathbf{B}$ and has the entries $a_{jk} + b_{jk}$ obtained by adding the corresponding entries of $\mathbf{A}$ and $\mathbf{B}$.

Matrices of different sizes **cannot be added**.

As a special case, the sum $\mathbf{a} + \mathbf{b}$ of two **row vectors** or two **column vectors**, which must have the same number of components, is obtained by adding the corresponding components.

Addition of Matrices and Vectors — 1.4 Example

If

$$\mathbf{A} = \begin{bmatrix} -4 & 6 & 3 \\ 0 & 1 & 2 \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 5 & -1 & 0 \\ 3 & 1 & 0 \end{bmatrix}, \quad \text{then} \quad \mathbf{A} + \mathbf{B} = \begin{bmatrix} 1 & 5 & 3 \\ 3 & 2 & 2 \end{bmatrix} \quad \blacksquare$$

$\mathbf{A}$ in Example 1.3 and our present $\mathbf{A}$ cannot be added.

Another example with vector as follows: $\mathbf{a} = \begin{bmatrix} 5 & 7 & 2 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} -6 & 2 & 0 \end{bmatrix}$, then $\mathbf{a} + \mathbf{b} = \begin{bmatrix} -1 & 9 & 2 \end{bmatrix}$.

Scalar Multiplication — 1.3 Definition

The **product** of any $m \times n$ matrix $\mathbf{A} = [a_{jk}]$ and any **scalar** c is written $c\mathbf{A}$ and is the $m \times n$ matrix $c\mathbf{A} = [ca_{jk}]$ obtained by multiplying each entry of $\mathbf{A}$ by c.

Here $(-1)\mathbf{A}$ is simply written $-\mathbf{A}$ and is called the **negative** of $\mathbf{A}$. Similarly, $(-k)\mathbf{A}$ is written written $-k\mathbf{A}$. In addition, $\mathbf{A} + (-\mathbf{B})$ is written $\mathbf{A} - \mathbf{B}$ and is called the **difference** of $\mathbf{A}$ and $\mathbf{B}$.

Scalar Multiplication — 1.5 Example

If

$$\mathbf{A} = \begin{bmatrix} 2.7 & -1.8 \\ 0 & 0.9 \\ 9.0 & -4.5 \end{bmatrix}, \quad \text{then} \quad -\mathbf{A} = \begin{bmatrix} -2.7 & 1.8 \\ 0 & -0.9 \\ -9.0 & 4.5 \end{bmatrix}, \quad \frac{10}{9}\mathbf{A} = \begin{bmatrix} 3 & -2 \\ 0 & 1 \\ 10 & -5 \end{bmatrix}, \quad 0\mathbf{A} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

If a matrix $\mathbf{B}$ shows the distances between some cities in miles, then multiplying that matrix with 1,609 (i.e., 1,609$\mathbf{B}$) would give these distances in kilometers.

Rules for Matrix Addition and Scalar Multiplication From the familiar laws for the addition of numbers we obtain similar laws for the addition of matrices of the same size $m \times n$, namely,

i. $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$

ii. $(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C})$ (written $\mathbf{A} + \mathbf{B} + \mathbf{C}$)

iii. $\mathbf{A} + \mathbf{0} = \mathbf{A}$

iv. $\mathbf{A} + (-\mathbf{A}) = \mathbf{0}$.

Here **0** denotes the **zero matrix**, of size $m \times n$,, that is, the $m \times n$ matrix with all entries zero. If $m = 1$ or $n = 1$, this is a vector, called a **zero vector**. Therefore matrix addition is **commutative** and **associative**. Similarly, for scalar multiplication we obtain the rules

$$\begin{aligned} &\text{i.} && c(\mathbf{A}+\mathbf{B}) = c\mathbf{A} + c\mathbf{B} \\ &\text{ii.} && (c+k)\mathbf{A} = c\mathbf{A} + k\mathbf{A} \\ &\text{iii.} && c(k\mathbf{A}) = (ck)\mathbf{A} \qquad \text{(written } ck\mathbf{A}\text{)} \\ &\text{iv.} && 1\mathbf{A} = \mathbf{A} \end{aligned}$$

1.1.4 Matrix Multiplication

Matrix multiplication means that one multiplies matrices by matrices. Its definition is standard but may looks artificial and slightly counter-intuitive. Therefore, here, we have to study matrix multiplication carefully.

We wills start, of course, with its definition.

Multiplication of a Matrix by a Matrix 1.4 Definition

The **product C = AB**, in this specific order, of an $m \times n$ matrix $\mathbf{A} = [a_{jk}]$ times an $r \times p$ matrix $\mathbf{B} = [b_{jk}]$ is defined if and only if $r = n$ and is then the $m \times p$ matrix $\mathbf{C} = [c_{ik}]$ with entries

$$c_{jk} = \sum_{l=1}^{n} a_{jl}b_{lk} = a_{j1}b_{1k} + a_{j2}b_{2k} + \cdots + a_{jn}b_{nk} \quad \text{where} \quad \begin{cases} j & = 1, \ldots, m \\ k & = 1, \ldots, p. \end{cases} \tag{1.3}$$

The condition $r = n$, which was given in Definition 1.4, means the 2nd factor, **B**, must have as many rows as the first factor has columns, namely n. A diagram of sizes that shows when matrix multiplication is possible is as follows:

$$\begin{matrix} \mathbf{A} & \mathbf{B} = \mathbf{C} \\ [m \times n] & [n \times p] = [m \times p]. \end{matrix}$$

The entry c_{ik} in Eq. (1.3) is obtained by multiplying each entry in the jth row of **A** by the corresponding entry in the kth column of **B** and then adding these n products.

For instance, $c_{21} = a_{21}b_{11} + a_{22}b_{21} + \cdots + a_{2n}b_{n1}$, and so on.

One calls this briefly a **multiplication** of rows into columns. For $n = 3$, this is illustrated in Figure 1.1 where we shaded the entries which contribute to the calculation of entry c_{21} just discussed.

Matrix multiplication will be motivated by its use in **linear transformations** in this section and more fully in Section 1.6.

Let's illustrate the primary points of matrix multiplication by some examples.

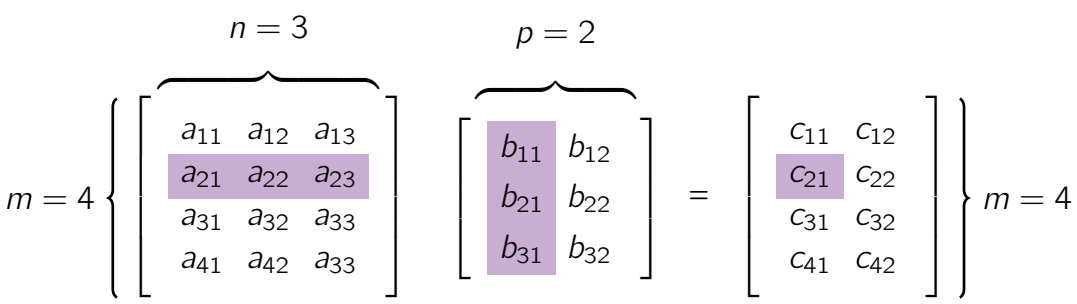


Figure 1.1
Notations in a product
AB = C

Matrix multiplication also includes multiplying a matrix by a vector, given a vector is a **special type of matrix**.

Matrix Multiplication — 1.6 Example

$$\mathbf{AB} = \begin{bmatrix} 3 & 5 & -1 \\ 4 & 0 & 2 \\ -6 & -3 & 2 \end{bmatrix} \begin{bmatrix} 2 & -2 & 3 & 1 \\ 5 & 0 & 7 & 8 \\ 9 & -4 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 22 & -2 & 43 & 42 \\ 26 & -16 & 14 & 6 \\ -9 & 4 & -37 & -28 \end{bmatrix}$$

Here $c_{11} = 3 \cdot 2 + 5 \cdot 5 + (-1) \cdot 9 = 22$, and so on. The entry in the box is $c_{23} = 4 \cdot 3 + 0 \cdot 7 + 2 \cdot 1 = 14$.

The product **BA** is not defined ■

Multiplication of a Matrix and a Vector — 1.7 Example

$$\begin{bmatrix} 4 & 2 \\ 1 & 8 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 4 \cdot 3 + 2 \cdot 5 \\ 1 \cdot 3 + 8 \cdot 5 \end{bmatrix} = \begin{bmatrix} 22 \\ 43 \end{bmatrix} \quad \text{whereas} \quad \begin{bmatrix} 3 \\ 5 \end{bmatrix} \begin{bmatrix} 4 & 2 \\ 1 & 8 \end{bmatrix} \quad \text{is undefined.}$$

Let's have a look at the product of row and column vectors:

$$\begin{bmatrix} 3 & 6 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 19 \end{bmatrix}, \qquad \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} \begin{bmatrix} 3 & 6 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 6 & 1 \\ 6 & 12 & 2 \\ 12 & 24 & 4 \end{bmatrix} \quad ■$$

Matrix Multiplication is Not Commutative — 1.8 Example

This is illustrated by Example 1.6 and first half of Example 1.7, where one of the two products is **NOT** even defined, and by second half of Example 1.7, where the two (**2**) products have different sizes. But it also holds for square matrices.

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For instance,

$$\begin{bmatrix} 1 & 1 \\ 100 & 100 \end{bmatrix}\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{but} \quad \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 100 & 100 \end{bmatrix} = \begin{bmatrix} 99 & 99 \\ -99 & -99 \end{bmatrix} \blacksquare$$

Interesting, as this also implies that $\mathbf{AB} = 0$ does **NOT** necessarily imply $\mathbf{BA} = 0$ or $\mathbf{A} = 0$ or $\mathbf{B} = 0$.

Our aforementioned examples show that in matrix **products the order of factors must always be observed very carefully**. Otherwise matrix multiplication satisfies rules similar to those for numbers, namely.

$$\begin{aligned} &\text{i.} && (k\mathbf{A})\mathbf{B} = k(\mathbf{AB}) = \mathbf{A}(k\mathbf{B}) && \text{written} \quad k\mathbf{AB} \quad \text{or} \quad \mathbf{A}k\mathbf{B} \\ &\text{ii.} && \mathbf{A}(\mathbf{BC}) = (\mathbf{AB})\mathbf{C} && \text{written } \mathbf{ABC} \\ &\text{iii.} && (\mathbf{A}+\mathbf{B})\mathbf{C} = \mathbf{AC} + \mathbf{BC} \\ &\text{iv.} && \mathbf{C}(\mathbf{A}+\mathbf{B}) = \mathbf{CA} + \mathbf{CB} \end{aligned} \tag{1.4}$$

provided $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ are such that the expressions on the left are defined; here, k is any scalar. Eq. (1.4)ii is called the **associative law**. Eq. (1.4)iii and Eq. (1.4)iv are called the **distributive laws**.

As matrix multiplication is a multiplication of rows into columns, we can write the defining formula Eq. (1.3) more compactly as

$$c_{jk} = \mathbf{a}_j\mathbf{b}_k, \qquad \text{where} \qquad j = 1, \ldots, m; \quad \text{and} \quad k = 1, \ldots, p, \tag{1.5}$$

where a_j is the jth row vector of $\mathbf{A}$ and b_k is the kth column vector of $\mathbf{B}$, so that in agreement with Eq. (1.3),

$$\mathbf{a}_j\mathbf{b}_k = \begin{bmatrix} a_{j1} & a_{j2} & \cdots & a_{jn} \end{bmatrix}\begin{vmatrix} b_{1k} \\ \vdots \\ b_{nk} \end{vmatrix} = a_{j1}b_{1k} + a_{j2}b_{2k} + \cdots + a_{jn}b_{nk}.$$

1.9 Example

Product in Terms of Row and Column Vectors

If $\mathbf{A} = [a_{jk}]$ is of size 3×3 and $\mathbf{B} = [b_{jk}]$ is of size 3×4, then

$$\mathbf{AB} = \begin{bmatrix} a_1b_1 & a_1b_2 & a_1b_3 & a_1b_4 \\ a_2b_1 & a_2b_2 & a_2b_3 & a_2b_4 \\ a_3b_1 & a_3b_2 & a_3b_3 & a_3b_4 \end{bmatrix} \blacksquare$$

1.1 Information

Parallel Processing of Products on Computers

This is facilitated by a variant of Eq. (1.5) for computing $\mathbf{C} = \mathbf{AB}$, which is used by standard algorithms.[9] In this method, $\mathbf{A}$ is used as given, $\mathbf{B}$ is taken in terms of its column vectors, and the product is computed columnwise;

$$\mathbf{AB} = \mathbf{A}[b_1 \quad b_2 \quad \cdots \quad b_p] = [Ab_1 \quad Ab_2 \quad \cdots \quad Ab_p].$$

[9] Such as LAPACK.

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Columns of **B** are then assigned to different processors,[10] which simultaneously compute the columns of the product matrix $\mathbf{Ab}_1$, $\mathbf{Ab}_2$ etc.

[10] individually or several to each processor.

Computing Products Columnwise — 1.10 Example

To obtain

$$\mathbf{AB} = \begin{bmatrix} 4 & 1 \\ -5 & 2 \end{bmatrix} \begin{bmatrix} 3 & 0 & 7 \\ -1 & 4 & 6 \end{bmatrix} = \begin{bmatrix} 11 & 4 & 34 \\ -17 & 8 & -23 \end{bmatrix}$$

from (5), calculate the columns

$$\begin{bmatrix} 4 & 1 \\ -5 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 11 \\ -17 \end{bmatrix}, \begin{bmatrix} 4 & 1 \\ -5 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 4 \end{bmatrix} = \begin{bmatrix} 4 \\ 8 \end{bmatrix}, \begin{bmatrix} 4 & 1 \\ -5 & 2 \end{bmatrix} \begin{bmatrix} 7 \\ 6 \end{bmatrix} = \begin{bmatrix} 34 \\ -23 \end{bmatrix}$$

of AB and then write them as a single matrix, as shown in the first formula on the right.

1.1.5 The use of Linear Transformations in Multiplication of Matrices

Let us now motivate the **unnatural** matrix multiplication by its use in **linear transformations.** For $n = 2$ variables these transformations are of the form:

$$\begin{aligned} y_1 &= a_{11}x_1 + a_{12}x_2 \\ y_2 &= a_{21}x_1 + a_{22}x_2 \end{aligned} \tag{1.6}$$

and suffice to explain the idea.[11] For example, Eq. (1.6) may relate an x_1x_2-coordinate system to a y_1y_2-coordinate system in the plane. In vectorial form we can write Eq. (1.6) as

[11] The general case will be discussed in Section 1.6.

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \mathbf{Ax} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 \\ a_{21}x_1 + a_{22}x_2 \end{bmatrix}. \tag{1.7}$$

Now suppose further that the x_1x_2-system is related to a w_1w_2-system by another linear transformation, say,

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \mathbf{Bw} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} b_{11}w_1 + b_{12}w_2 \\ b_{21}w_1 + b_{22}w_2 \end{bmatrix}. \tag{1.8}$$

Then the y_1y_2-system is related to the w_1w_2-system indirectly via the x_1x_2-system, and we wish to express this relation directly. Substitution will show that this direct relation is a linear transformation, too, say,

$$\mathbf{y} = \mathbf{Cw} = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} c_{11}w_1 + c_{12}w_2 \\ c_{21}w_1 + c_{22}w_2 \end{bmatrix}. \tag{1.9}$$

Indeed, substituting Eq. (1.8) into Eq. (1.7), we obtain

$$\begin{aligned} y_1 &= a_{11}(b_{11}w_1 + b_{12}w_2) + a_{12}(b_{21}w_1 + b_{22}w_2) \\ &= (a_{11}b_{11} + a_{12}b_{21})w_1 + (a_{11}b_{12} + a_{12}b_{22})w_2 \\ y_2 &= a_{21}(b_{11}w_1 + b_{12}w_2) + a_{22}(b_{21}w_1 + b_{22}w_2) \\ &= (a_{21}b_{11} + a_{22}b_{21})w_1 + (a_{21}b_{12} + a_{22}b_{22})w_2 \end{aligned}$$

Comparing this with Eq. (1.9), we see that

$$c_{11} = a_{11}b_{11} + a_{12}b_{21}$$
$$c_{12} = a_{11}b_{12} + a_{12}b_{22}$$
$$c_{21} = a_{21}b_{11} + a_{22}b_{21}$$
$$c_{22} = a_{21}b_{12} + a_{22}b_{22}.$$

This proves that $\mathbf{C} = \mathbf{AB}$ with the product defined as in Eq. (1.3). For larger matrix sizes the idea and result are exactly the same. Only the number of variables changes.

We then have m variables y and n variables x and p variables w.

The matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C} = \mathbf{AB}$ then have sizes $m \times n$, $n \times p$, and $m \times p$, respectively. And the requirement that $\mathbf{C}$ be the product $\mathbf{AB}$ leads to formula Eq. (1.3) in its general form.

Transposition

We obtain the **transpose**[12] of a matrix by writing its rows as columns, or equivalently its columns as rows. This also applies to the transpose of vectors. Therefore, a row vector becomes a column vector and vice versa.

[12] comes from the Old French *transposer*, to transfer or remove.

In addition, for square matrices, we can also **reflect** the elements along the main diagonal, that is, interchange entries that are symmetrically positioned with respect to the main diagonal to obtain the transpose.

Hence a_{12} becomes a_{21}, a_{31} becomes a_{13}, and so forth.

The Example 1.11 illustrates these ideas. Also note that, if $\mathbf{A}$ is the given matrix, then we denote its transpose by $\mathbf{A}^T$.

1.11 Example

Transposition of Matrices and Vectors

If

$$\mathbf{A} = \begin{bmatrix} 5 & -8 & 1 \\ 4 & 0 & 0 \end{bmatrix}, \quad \text{then} \quad \mathbf{A}^T = \begin{bmatrix} 5 & 4 \\ -8 & 0 \\ 1 & 0 \end{bmatrix}.$$

A little more compactly, we can write

$$\begin{bmatrix} 5 & -8 & 1 \\ 4 & 0 & 0 \end{bmatrix}^T = \begin{bmatrix} 5 & 4 \\ -8 & 0 \\ 1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 3 & 0 & 7 \\ 8 & -1 & 5 \\ 1 & -9 & 4 \end{bmatrix}^T = \begin{bmatrix} 3 & 8 & 1 \\ 0 & -1 & -9 \\ 7 & 5 & 4 \end{bmatrix},$$

Furthermore, the transpose $\begin{bmatrix} 6 & 2 & 3 \end{bmatrix}^T$ of the row vector $\begin{bmatrix} 6 & 2 & 3 \end{bmatrix}$ is the column vector

$$\begin{bmatrix} 6 & 2 & 3 \end{bmatrix}^T = \begin{bmatrix} 6 \\ 2 \\ 3 \end{bmatrix}. \quad \text{Conversely,} \quad \begin{bmatrix} 6 \\ 2 \\ 3 \end{bmatrix}^T = \begin{bmatrix} 6 & 2 & 3 \end{bmatrix}.$$

Transposition of Matrices and Vectors 1.5 Definition

The transpose of an $m \times n$ matrix $\mathbf{A} = [a_{jk}]$ is the $n \times m$ matrix $\mathbf{A}^T$[13] that has the first row of $\mathbf{A}$ as its first column, the second row of $\mathbf{A}$ as its second *column*, and so on. Therefore, the transpose of $\mathbf{A}$ in Eq. (1.4) is $\mathbf{A}^T = [a_{kj}]$, written out

$$\mathbf{A}^T = [a_{kj}] = \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \ddots & \ddots \end{bmatrix}.$$

As a special case, transposition converts row vectors to column vectors and conversely.

[13] This is read as A transpose

Transposition gives us a choice in that we can work either with the matrix or its transpose, whichever is more convenient. Rules for transposition are

$$\begin{aligned} &\text{i.} && (\mathbf{A}^T)^T = \mathbf{A} \\ &\text{ii.} && (\mathbf{A} + \mathbf{B})^T = \mathbf{A}^T + \mathbf{B}^T \\ &\text{iii.} && (c\mathbf{A})^T = c\mathbf{A}^T \\ &\text{iv.} && (\mathbf{AB})^T = \mathbf{B}^T\mathbf{A}^T. \end{aligned}$$

Note the last equation where the transposed matrices are **in reversed order**.

Special Matrices

Certain kinds of matrices will occur quite frequently in our work as engineers, and we now list the most important ones of them to hopefully get familiarity.

Symmetric and Skew-Symmetric Matrices Transposition gives rise to two (2) useful classes of matrices.

i. Symmetric matrices are square matrices whose transpose equals the matrix itself.

ii. Skew-symmetric matrices are square matrices whose transpose equals minus the matrix.

Both cases are defined in Eq. (1.10) and illustrated by Example 8.

$$\begin{aligned} &\mathbf{A}^T = \mathbf{A} \quad \text{therefore} \quad a_{kj} = a_{jk}, \\ &\mathbf{A}^T = -\mathbf{A} \quad \text{therefore} \quad a_{kj} = -a_{jk}, \quad \text{and} \quad a_{jj} = 0. \end{aligned} \tag{1.10}$$

Symmetric Matrix and Skew-Symmetric Matrix — 1.12 Example

$$\mathbf{A} = \begin{bmatrix} 20 & 120 & 200 \\ 120 & 10 & 150 \\ 200 & 150 & 30 \end{bmatrix} \text{ is symmetric} \qquad \mathbf{B} = \begin{bmatrix} 0 & 1 & -3 \\ -1 & 0 & -2 \\ 3 & 2 & 0 \end{bmatrix} \text{ is skew-symmetric.}$$

Triangular Matrices Upper triangular matrices are square matrices which can have nonzero entries only on and above the main diagonal, whereas any entry below the diagonal must be zero. Similarly, lower triangular matrices can have nonzero entries only on and below the main diagonal. Any entry on the main diagonal of a triangular matrix may be zero or not.

Upper and Lower Triangular Matrices — 1.13 Example

The following are example matrices for upper and lower triangular matrices.

$$\begin{bmatrix} 1 & 3 \\ 0 & 2 \end{bmatrix}, \begin{bmatrix} 1 & 4 & 2 \\ 0 & 3 & 2 \\ 0 & 0 & 6 \end{bmatrix} \quad \begin{bmatrix} 2 & 0 & 0 \\ 8 & -1 & 0 \\ 7 & 6 & 8 \end{bmatrix}, \begin{bmatrix} 3 & 0 & 0 & 0 \\ 9 & -3 & 0 & 0 \\ 1 & 0 & 2 & 0 \\ 1 & 0 & 3 & 6 \end{bmatrix}.$$

Upper Triangular — Lower Triangular

Diagonal Matrices These are square matrices which can have nonzero entries **only** on the main diagonal.

> Any entry above or below the main diagonal must be zero.

If all the diagonal entries of a diagonal matrix $\mathbf{S}$ are equal, say, c, we call $\mathbf{S}$ a scalar matrix because multiplication of any square matrix $\mathbf{A}$ of the same size by $\mathbf{S}$ has the same effect as the multiplication by a scalar, that is,

$$\mathbf{AS} = \mathbf{SA} = c\mathbf{A} \tag{1.11}$$

In particular, a scalar matrix, whose entries on the main diagonal are all 1, is called a **unit** matrix[14] and is denoted by $\mathbf{I}_n$ or simply by $\mathbf{I}$. For $\mathbf{I}$, formula Eq. (1.11) becomes:

$$\mathbf{AI} = \mathbf{IA} = \mathbf{A}$$

[14] This is also known as an **identity** matrix.

Examples of Diagonal Matrices — 1.14 Example

Some examples of diagonal matrices are shown below:

$$\mathbf{D} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{S} = \begin{bmatrix} c & 0 & 0 \\ 0 & c & 0 \\ 0 & 0 & c \end{bmatrix}, \quad \mathbf{I} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \blacksquare$$

1.2 Linear Systems of Equations Gauss Elimination

We now come to one of the most important use of matrices, that is, using matrices for **solving systems of linear equations**. We showed informally, in Example 1.1 of Section 1.1, how to represent the information contained in a system of linear equations by a matrix, called the **augmented matrix**. This matrix will then be used in solving the linear system of equations. Our approach to solving linear systems is called the **Gauss elimination method**.

This method is fundamental to linear algebra, and therefore should be understood thoroughly.

A shorter term for systems of linear equations is just linear systems. Linear systems model many applications in engineering, economics, statistics, and many other areas. Electrical networks, traffic flow, and commodity markets may serve as specific examples of applications.

1.2.1 Linear System, Coefficient Matrix, Augmented Matrix

A linear system of m equations in n unknowns $x_1, \ldots, x_n$ is a set of equations of the form:

$$\begin{aligned} a_{11}x_1 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + \cdots + a_{2n}x_n &= b_2 \\ \cdots &\vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n &= b_m. \end{aligned} \tag{1.12}$$

The system is called **linear** given each variable x_j appears in the first power only, just as in the equation of a straight line. $a_{11}, \ldots, a_{mn}$ are given numbers, called the **coefficients** of the system. $b_1, \ldots, b_m$ on the right are also given numbers. If all the b_j are zero, then Eq. (1.12) is called a **homogeneous system**. If at least one b_j is not zero, then Eq. (1.12) is called a **nonhomogeneous** system.

A **solution** of Eq. (1.12) is a set of numbers $x_1, \ldots, x_n$ that satisfies all the m equations. A solution vector of Eq. (1.12) is a vector x whose components form a solution of Eq. (1.12). If the system Eq. (1.12) is homogeneous, it always has at least the **trivial solution** $x_1 = 0, \ldots, x_n = 0$.

Matrix Form of Linear Systems

From the definition of matrix multiplication we see that the m equations of Eq. (1.12) may be written as a single vector equation:

$$\mathbf{Ax} = \mathbf{b}$$

where the **coefficient matrix A** $= [a_{ik}]$ is the $m \times n$ matrix

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad \text{and} \quad \mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

are column vectors. We assume that the coefficients a_{ik} are not all zero, so that **A** is not a zero matrix. Note that **x** has n components, whereas **b** has m components. The matrix

$$\tilde{\mathbf{A}} = \left[\begin{array}{ccc|c} a_{11} & \cdots & a_{1n} & b_1 \\ \vdots & \cdots & \vdots & \vdots \\ \vdots & \cdots & \vdots & \ddots \\ & \ddots & \ddots & \ddots \\ & \ddots & \ddots & \ddots \end{array}\right]$$

is called the **augmented matrix** of the system Eq. (1.12). The vertical line could be omitted, as we will do later.

It is merely a reminder that the last column of **A** did not come from matrix **A** but came from vector **b**.

Therefore, we **augmented** the matrix **A**. Note that the augmented matrix **A** determines the system Eq. (1.12) completely because it contains all the given numbers appearing in Eq. (1.12).

1.15 Example

Geometric Interpretation. Existence and Uniqueness of Solutions

If $m = n = 2$, we have two (2) equations in two (2) unknowns x_1, x_2, namely:

$$a_{11}x_1 + a_{12}x_2 = b_1$$
$$a_{21}x_1 + a_{22}x_2 = b_2.$$

If we interpret x_1, x_2 as coordinates in the x_1x_2-plane, then each of the two equations represents a straight line, and (x_1, x_2) is a solution if and only if the point P with coordinates x_1, x_2 lies on both lines. Hence there are three (3) possible cases:

1. Precisely one solution if the lines intersect
2. Infinitely many solutions if the lines coincide
3. No solution if the lines are parallel

If the system is homogenous, Case 1. cannot happen, because then those two straight lines pass through the origin, whose coordinates (0, 0) constitute the trivial solution.

Our simple aforementioned example illustrated that a system given in Eq. (**1.12**) may have no solution. This leads to such questions as:

> Does a given system in Eq. (**1.12**) have a solution? Under what conditions does it have precisely one solution? If it has more than one solution, how can we characterize the set of all solutions?

First, before looking at this in significant detail, let us discuss an important systematic method for solving linear systems.

1.2.2 Gauss Elimination and Back Substitution

The Gauss elimination method can be explained as follows.[15]

Consider a linear system that is in triangular form[16] such as:

$$2x_1 + 5x_2 = 2 \qquad \text{and} \qquad 13x_2 = -26$$

> Triangular means all the **nonzero entries of the corresponding coefficient matrix lie above the diagonal** and form an upside-down 90° triangle.

Then we can solve the system by **back substitution**, that is, we solve the last equation for the variable, $x_2 = -26/13 = -2$, No solution and then work backward, substituting $x_2 = -2$ into the first equation and solving it for x_1, obtaining $x_1 = \frac{1}{2}(2 - 5x_2) = \frac{1}{2}(2 - 5 \cdot (-2)) = 6$. This gives us the idea of first reducing a general system to triangular form. For instance, let the given system be

$$\begin{array}{rcl} 2x_1 & + & 5x_2 = 2 \\ -4x_1 & + & 3x_2 = -30. \end{array} \qquad \text{where its augmented matrix is} \qquad \begin{bmatrix} 2 & 5 & 2 \\ -4 & 3 & -30 \end{bmatrix}.$$

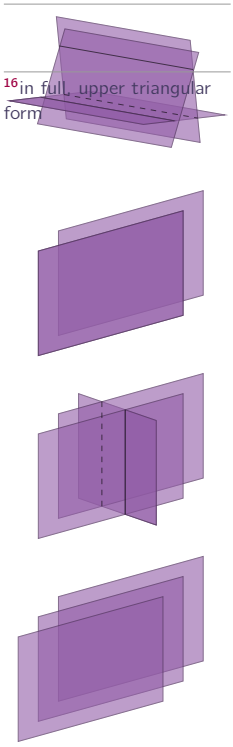
We leave the first equation as it is. We eliminate x_1 from the second equation, to get a triangular system. For this we add twice the first equation to the second, and we do the same operation on the **rows** of the augmented matrix.

This gives $-4x_1 + 4x_1 + 3x_2 + 10x_2 = -30 + 2 \cdot 2$, that is,

$$\begin{array}{rcl} 2x_1 & + & 5x_2 = 2 \\ & & 13x_2 = -26 \end{array} \qquad \text{Row 2 + 2 Row 1} \qquad \begin{bmatrix} 2 & 5 & 2 \\ 0 & 13 & -26 \end{bmatrix}$$

where **Row 2 + 2 Row 1** means *Add twice Row 1 to Row 2* in the original matrix. This is the Gauss elimination[17] giving the triangular form, from which back substitution now give us $x_2 = -2$ and $x_1 = 6$, as before.

Since a linear system is completely determined by its augmented matrix, **Gauss elimination** can be done by merely considering the matrices, as we have just indicated. We do this again in the next



[15] Examples of possible solutions a system of linear equations can take.

[16] in full, upper triangular form

[17] for 2 equations in 2 unknowns.

example, emphasizing the matrices by writing them first and the equations behind them, just as a help in order not to lose track.

Solving an Electrical Network 1.16 Example

Gauss Elimination. Electrical Network

1.2.3 Elementary Row Operations. Row-Equivalent Systems

The aforementioned Example 1.16 illustrates the operations of the Gauss elimination. These are the first two of three operations, which are called

Elementary Row Operations for Matrices The steps which are as follows:

- Interchange of two rows
- Addition of a constant multiple of one row to another row
- Multiplication of a row by a **nonzero** constant c.

These operations are for rows, **not for columns**.

Elementary Operations for Equations The steps which are as follows:

- Interchange of two equations
- Addition of a constant multiple of one equation to another equation
- Multiplication of an equation by a nonzero constant c.

Clearly, the interchange of two equations does not alter the solution set. Neither does their addition because we can undo it by a corresponding subtraction. Similarly for their multiplication, which we can undo by multiplying the new equation by $1/c$, since $c \neq 0$, producing the original equation.

We now call a linear system S_1 row-equivalent to a linear system S_2 if S_1 can be obtained from S_2 by finitely many row operations. This justifies Gauss elimination and establishes the following result.

Row-Equivalent Systems 1.1 Theorem

Row-equivalent linear systems have the same set of solutions.

Because of this theorem, systems having the same solution sets are often called **equivalent systems**. But note well that we are dealing with row operations. No column operations on the augmented

matrix are permitted in this context because they would generally alter the solution set.

A linear system given in Eq. (**1.12**) is called **overdetermined** if it has more equations than unknowns, as in Example 1.16, **determined** if $m = n$, as in Example 1.15, and **underdetermined** if it has fewer equations than unknowns.

In addition, a system of Eq. (**1.12**) is called **consistent** if it has at least one solution,[18] but **inconsistent** if it has no solutions at all, as $x_1 + x_2 = 1, x_1 + x_2 = 0$ in Example 1.15, Case c.

[18] Therefore, one solution or infinitely many solutions.

1.3 Linear Independence and Rank of a Matrix Vector Space

Given our next goal is to **fully characterize** the behavior of linear systems in terms of **existence** and **uniqueness** of solutions which is in Section 1.3, we have to introduce new fundamental linear algebraic concepts that will aid us in doing so. Foremost among these are **linear independence** and the **rank** of a matrix.c

These concepts are intimately linked with the important Gauss elimination method and how it works.

1.3.1 Linear Independence and Dependence of Vectors

Given any set of m vectors $\mathbf{a}_{(1)}, \ldots, \mathbf{a}_{(m)}$,[19] a linear combination of these vectors is an expression of the form:

$$c_1\mathbf{a}_{(1)} + c_2\mathbf{a}_{(2)} + \cdots + c_m\mathbf{a}_{(m)}$$

where $c_1, c_2, \ldots, c_m$ are any scalars. Now consider the equation

$$c_1\mathbf{a}_{(1)} + c_2\mathbf{a}_{(2)} + \cdots + c_m\mathbf{a}_{(m)} = \mathbf{0}. \tag{1.13}$$

Clearly, this vector equation Eq. (1.13) holds if we choose all c_j's zero, because then it becomes $\mathbf{0} = \mathbf{0}$. If this is the only m-tuple of scalars for which Eq. (1.13) holds, then our vectors $\mathbf{a}_{(1)}, \ldots, \mathbf{a}_{(m)}$ are said to form a **linearly independent set** or, more briefly, we call them **linearly independent**. Otherwise, if Eq. (1.13) also holds with scalars not all zero, we call these vectors **linearly dependent**. This means that we can express at least one of the vectors as a linear combination of the other vectors. For instance, if Eq. (1.13) holds with, say, $c_1 \neq 0$, we can solve Eq. (1.13) for $\mathbf{a}_{(1)}$:[20]

$$\mathbf{a}_{(1)} = k_2\mathbf{a}_{(2)} + \cdots + k_m\mathbf{a}_{(m)} \qquad \text{where} \qquad k_j = -c_j/c_1$$

But why is linear independence important?

Well, if a set of vectors is linearly dependent, then we can get rid of at least one or perhaps more of the vectors until we get a linearly independent set. This set is then the smallest **truly essential**x set with which we can work.

Thus, we cannot express any of the vectors, of this set, linearly in terms of the others.

Linear Independence and Dependence — 1.17 Example

The three vectors shown below:

$$\mathbf{a}_{(1)} = \begin{bmatrix} 3 & 0 & 2 & 2 \end{bmatrix}$$
$$\mathbf{a}_{(2)} = \begin{bmatrix} -6 & 42 & 24 & 54 \end{bmatrix}$$
$$\mathbf{a}_{(3)} = \begin{bmatrix} 21 & -21 & 0 & -15 \end{bmatrix}$$

Continues on next page →

[19] with the same number of components.

[20] Some k_j's may be zero. Or even all of them, namely, if $a_{(1)} = 0$.

← Continued from previous page

are **linearly dependent** as:

$$6\mathbf{a}_{(1)} - \frac{1}{2}\mathbf{a}_{(2)} - \mathbf{a}_{(3)} = \mathbf{0}.$$

Although this is easily checked by vector arithmetic, it is not so easy to discover. However, a systematic method for finding out about linear independence and dependence follows below.

First two vectors are linearly independent as $c_1\mathbf{a}_{(1)} + c_2\mathbf{a}_{(2)} = \mathbf{0}$ implies $c_2 = 0$ and $c_1 = 0$.

1.3.2 Rank of a Matrix

Before we begin, let's properly define what a rank is.

Rank of a Matrix — 1.6 Definition

The **rank** of a matrix $\mathbf{A}$ is the maximum number of linearly independent row vectors of $\mathbf{A}$. It is denoted by rank $\mathbf{A}$.

Our further discussion will show that the rank of a matrix is an important key concept for understanding general properties of matrices and linear systems of equations.

Rank of a Matrix — 1.18 Example

The matrix

$$\mathbf{A} = \begin{bmatrix} 3 & 0 & 2 & 2 \\ -6 & 42 & 24 & 54 \\ 21 & -21 & 0 & -15 \end{bmatrix}$$

has rank 2, as Example 1.17 shows that the first two row vectors are **linearly independent**, whereas all three row vectors are linearly dependent.

---

rank $\mathbf{A} = 0$ if and only if $\mathbf{A} = 0$. This follows directly from the definition.

We call a matrix $\mathbf{A}_1$ **row-equivalent** to a matrix $\mathbf{A}_2$ if $\mathbf{A}_1$ can be obtained from $\mathbf{A}_2$ by finitely many, elementary row operations.

Now the maximum number of linearly independent row vectors of a matrix does **NOT** change if we change the order of rows or multiply a row by a nonzero c or take a linear combination by adding a multiple of a row to another row.

This shows that rank is **invariant** under elementary row operations:

Row-Equivalent Matrices — 1.2 Theorem

Row-equivalent matrices have **the same rank**.

Therefore we can determine the rank of a matrix by reducing the matrix to row-echelon form, as was done in Section 1.2. Once the matrix is in row-echelon form, we count the number of nonzero rows, which is precisely the rank of the matrix.

Determination of Rank — 1.19 Example

For the matrix in Example 2 we obtain successively

$$\mathbf{A} = \begin{bmatrix} 3 & 0 & 2 & 2 \\ -6 & 42 & 24 & 54 \\ 21 & -21 & 0 & -15 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 0 & 2 & 2 \\ 0 & 42 & 28 & 58 \\ 0 & -21 & -14 & -29 \end{bmatrix} \quad \begin{matrix} \\ \text{Row } 2 + 2 \text{ Row } 1 \\ \text{Row } 3 - 7 \text{ Row } 1 \end{matrix}$$

$$\begin{bmatrix} 3 & 0 & 2 & 2 \\ 0 & 42 & 28 & 58 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{Row } 3 + \tfrac{1}{2}\text{Row } 2.$$

The last matrix is in row-echelon form and has two nonzero rows. Hence rank $\mathbf{A} = 2$, as before ■

Examples 13 illustrate the following useful theorem (with $p = 3$, $n = 3$, and the rank of the matrix $= 2$).

Linear Independence and Dependence of Vectors — 1.3 Theorem

Consider p vectors that each have n components. Then these vectors are **linearly independent** if the matrix formed, with these vectors as row vectors, has rank p.

However, these vectors are **linearly dependent** if that matrix has rank less than p.

Further important properties will result from the basic

Rank in Terms of Column Vectors — 1.4 Theorem

The rank r of a matrix $\mathbf{A}$ equals the maximum number of linearly independent column vectors of $\mathbf{A}$. Hence $\mathbf{A}$ and its transpose $\mathbf{A}^T$ have the same rank.

Illustration of "Rank in Terms of Column Vectors" — 1.20 Example

Illustration of Theorem 3 The matrix in has rank 2. From Example 1.19 we see that the first two row vectors are **linearly independent** and by *working backward* we can verify that:

$$\text{Row } 3 = 6 \text{ Row } 1 - \frac{1}{2} \text{ Row } 2$$

Similarly, the first two columns are linearly independent, and by reducing the last matrix in Example 1.19 by columns we find that:

$$\text{Column } 3 = \frac{2}{3}\text{Column } 1 + \frac{2}{3}\text{Column } 2 \qquad \text{and} \qquad \text{Column } 4 = \frac{2}{3}\text{Column } 1 + \frac{29}{21}\text{Column } 2 \quad \blacksquare$$

Combining Theory 1.3 and we obtain

Linear Dependence of Vectors — 1.5 Theorem

Consider p vectors each having n components. If $n < p$, then these vectors are **linearly dependent**.

1.3.3 Vector Space

The following related concepts are of general interest in **linear algebra**.

> In the present context they provide a clarification of essential properties of matrices and their role in connection with linear systems.

Consider a nonempty set V of vectors where each vector has the same number of components. If, for any two vectors $\mathbf{a}$ and $\mathbf{b}$ in V, we have that all their linear combinations $\alpha\mathbf{a} + \beta\mathbf{b}$, where α and β are any real numbers, are also elements of V, and if, furthermore, $\mathbf{a}$ and $\mathbf{b}$ satisfy the laws in (3a), (3c), (3d), and (4) in Sec. 7.1, as well as any vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ in V satisfy (3b) then V is a vector space.

Note that here we wrote laws (3) and (4) of Sec. 7.1 in lowercase letters a, b, c, which is our notation for vectors. More on vector spaces in Sec. 7.9. The maximum number of linearly independent vectors in V is called the **dimension** of V and is denoted by dim V. Here we assume the dimension to be finite; infinite dimension will be defined in Sec. 7.9.

A linearly independent set in V consisting of a maximum possible number of vectors in V is called a **basis** for V. In other words, any largest possible set of independent vectors in V forms basis for V.

> That means, if we add one or more vector to that set, the set will be linearly dependent.

Therefore, the number of vectors of a basis for V equals dim V.

The set of all linear combinations of given vectors $\mathbf{a}_{(1)}, \ldots, \mathbf{a}_{(p)}$ with the same number of components is called the **span** of these vectors. Obviously, a span is a vector space. If in addition, the given

vectors $\mathbf{a}_{(1)}, \ldots, \mathbf{a}_{(p)}$ are linearly independent, then they form a basis for that vector space.

This then leads to another equivalent definition of basis. A set of vectors is a basis for a vector space V if (1) the vectors in the set are linearly independent, and if (2) any vector in V can be expressed as a linear combination of the vectors in the set. If (2) holds, we also say that the set of vectors **spans** the vector space V.

By a **subspace** of a vector space V we mean a nonempty subset of V (including V itself) that forms a vector space with respect to the two algebraic operations (addition and scalar multiplication) defined for the vectors of V.

1.21 Example

Vector Space, Dimension, and Basis

The span of the three vectors in Example 1 is a vector space of dimension 2. A basis of this vector space consists of any two of those three vectors, for instance, $\mathbf{a}_{(1)}$, $\mathbf{a}_{(2)}$, or $\mathbf{a}_{(1)}$, $\mathbf{a}_{(3)}$, etc.

We further note the following theorem

1.6 Theorem

Vector Space

The vector space $\mathbb{R}^n$ consisting of all vectors with n components has dimension n.

For a matrix $\mathbf{A}$, we call the span of the row vectors the **row space** of $\mathbf{A}$. Similarly, the span of the column vectors of $\mathbf{A}$ is called the **column space** of $\mathbf{A}$. Now, Theorem 3 shows that a matrix A has as many linearly independent rows as columns. By the definition of dimension, their number is the dimension of the row space or the column space of $\mathbf{A}$. This proves the following:

1.7 Theorem

Row Space and Column Space

The row space and the column space of a matrix A have the same dimension, equal to rank A.

Finally, for a given matrix $\mathbf{A}$ the solution set of the homogeneous system $\mathbf{Ax} = 0$ is a vector space, called the **null space** of $\mathbf{A}$, and its dimension is called the **nullity** of $\mathbf{A}$ with the following important relation:

$$rank A + nullity A = Number of columns of A.$$

1.3.4 Solutions of Linear Systems: Existence, Uniqueness

Rank, as defined in Section 1.3, gives complete information about existence, uniqueness, and general structure of the solution set of linear systems as follows. A linear system of equations in n unknowns has a unique solution if the coefficient matrix and the augmented matrix have the same rank n, and infinitely many solutions if that common rank is less than n.

The system has no solution if those two matrices have different rank.

To state this precisely and prove it, we shall use the generally important concept of a submatrix of A. By this we mean any matrix obtained from A by omitting some rows or columns (or both). By definition this includes A itself (as the matrix obtained by omitting no rows or columns); this is practical.

Fundamental Theorem for Linear Systems — 1.8 Theorem

i. **Existence** A linear system of m equations in n unknowns $x_1, \ldots, x_n$

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ \cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots & \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned} \tag{1.14}$$

is **consistent**, which means it has solutions, if and only if the coefficient matrix **A** and the augmented matrix $\tilde{\mathbf{A}}$ have the **same rank**. Here,

ii. **Uniqueness** The system given in Eq. (1.1) has precisely one solution if and only if this common rank r of A and $\tilde{A}$ equals n.

iii. **Infinitely many solutions** If this common rank r is less than n, the system Eq. (1.1) has infinitely many solutions. All of these solutions are obtained by determining r suitable unknowns[21] in terms of the remaining $n - r$ unknowns, to which arbitrary values can be assigned.

iv. **Gauss elimination** If solutions exist, they can all be obtained by the Gauss elimination.

[21] whose submatrix of coefficients must have rank r.

The theorem is illustrated in Section 1.2. In Example 2 there is a unique solution since rank A = rank $A = n = 3$ (as can be seen from the last matrix in the example). In Example 3 we have rank $\mathbf{A}$ = rank $\mathbf{A} = 2 < n = 4$ and can choose x_3 and x_4 arbitrarily. In Example 4 there is no solution because rank $\mathbf{A} = 2 <$ rank $\tilde{\mathbf{A}} = 3$.

Homogeneous Linear System

Recall from Section 1.2 that a linear system given in Eq. (1.14) is called **homogeneous** if all the b_j's are zero, and **nonhomogeneous** if one or several b_i's are not zero. For the homogeneous system we obtain from the Fundamental Theorem the following results.

Homogeneous Linear System — 1.9 Theorem

A homogeneous linear system of the form:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= 0 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= 0 \\ &\cdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= 0 \end{aligned} \tag{1.15}$$

always has the **trivial solution** $x_1 = 0, \ldots, x_n = 0$. Nontrivial solutions exist if and only if rank $A < n$. If rank $A = r < n$, these solutions, together with $x = 0$, form a vector space (see Sec. 7.4) of dimension $n - r$ called the solution space of (4).

In particular, if $\mathbf{x}_{(1)}$ and $\mathbf{x}_{(2)}$ are solution vectors of (4), then $\mathbf{x} = c_1\mathbf{x}_{(1)} + c_2\mathbf{x}_{(2)}$ with any scalars c_1 and c_2 is a solution vector of Eq. (1.15).[22]

[22] This **does not hold** for non-homogeneous systems. Also, the term solution space is used for homogeneous systems only.

The solution space of Eq. (1.15) is also called the **null space** of $\mathbf{A}$ as $\mathbf{Ax} = \mathbf{0}$ for every $\mathbf{x}$ in the solution space of (4). Its dimension is called the **nullity** of $\mathbf{A}$. Therefore Theory 1.9 states that

$$(5) rank A + nullity A = n$$

where n is the number of unknowns.[23] Furthermore, by the definition of rank we have rank $A \leqslant m$ in (4). Hence if $m < n$, then rank $\mathbf{A} < n$. By Theory 1.9, this gives the practically important

[23] Or we can say it is the numbers of columns of A.

Homogeneous Linear System with Fewer Equations Than Unknowns — 1.10 Theorem

A homogeneous linear system with fewer equations than unknowns **always has nontrivial solutions**.

Nonhomogeneous Linear Systems

The characterization of all solutions of the linear system (1) is now quite simple, as follows.

Nonhomogeneous Linear System — 1.11 Theorem

If a nonhomogeneous linear system (1) is consistent, then all of its solutions are obtained as

$$\mathbf{x} = \mathbf{x}_0 + \mathbf{x}_h$$

where $\mathbf{x}_0$ is any (fixed) solution of (1) and $\mathbf{x}_h$ runs through all the solutions of the corresponding homogeneous system (4). The difference $\mathbf{x}_h = \mathbf{x} - \mathbf{x}_0$ of any two solutions of (1) is a solution of (4) because

This covers a main part of our discussion of characterizing the solutions of systems of linear equations. Our next main topic is determinants and their role in linear equations.

1.4 Determinants and Cramer's Rule

Determinants[24] were originally introduced for solving linear systems and therefore has a significant place in their application to wide variety of engineering problems. However, before we tackle them in detail, it is worthwhile to have a good refreshment on the topic.

[24] Historically, determinants were used long before matrices: A determinant was originally defined as a property of a system of linear equations. The determinant "determines" whether the system has a unique solution, which occurs precisely if the determinant is non-zero. In Europe, solutions of linear systems of two equations were expressed by *Cardano* in 1545 by a determinant-like entity [1].

1.4.1 A Quick Refreshment on Determinants

A **determinant of second order** is denoted and defined by

$$D = \det \mathbf{A} = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}, \tag{1.16}$$

where we use bars instead of brackets to denote the aforementioned operation is a **determinant**.

Cramer's Rule for solving linear systems of two (2) equations in two unknowns is represented as follows:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 &= b_1 \\ a_{21}x_1 + a_{22}x_2 &= b_2 \end{aligned} \qquad \text{where} \qquad \begin{aligned} x_1 &= \frac{\begin{vmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{vmatrix}}{D} = \frac{b_1 a_{22} - a_{12} b_2}{D}, \\ x_2 &= \frac{\begin{vmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{vmatrix}}{D} = \frac{a_{11} b_2 - b_1 a_{21}}{D} \end{aligned}$$

with D as in Eq. (1.16), provided that $D \neq 0$, where the value $D = 0$ appears for homogeneous systems with nontrivial solutions.

1.22 Example

Cramer's Rule for Two Equation

If

$$\begin{aligned} 4x_1 + 3x_2 &= 12 \\ 2x_1 + 5x_2 &= -8 \end{aligned}$$

then we can calculate the unknowns as:

$$x_1 = \begin{vmatrix} 12 & 3 \\ -8 & 5 \end{vmatrix} = \frac{84}{14} = 6, \quad x_2 = \begin{vmatrix} 4 & 12 \\ 2 & -8 \end{vmatrix} = \frac{-56}{14} = -4.$$

1.4.2 Third-Order Determinants

A **determinant of third order** can be defined by as follows:

$$D = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{21} \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} + a_{31} \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix} \tag{1.17}$$

Note the following. The signs on the right are $+ - +$. Each of the three terms on the right is an entry in the first column of D times its **minor**, that is, the second-order determinant obtained from D by deleting the row and column of that entry; therefore, for a_{11} delete the first row and first column, and so on. If we write out the minors in Eq. (**1.17**), we obtain:

$$D = a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32} + a_{21}a_{13}a_{32} - a_{21}a_{12}a_{33} + a_{31}a_{12}a_{23} - a_{31}a_{13}a_{22}. \tag{1.18}$$

Cramer's Rule for Linear Systems of Three Equations

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3 \end{aligned} \quad \text{is} \quad x_1 = \frac{D_1}{D}, \quad x_2 = \frac{D_2}{D}, \quad x_3 = \frac{D_3}{D} \quad \text{where} \quad (D \neq 0) \tag{1.19}$$

with the determinant D of the system given by Eq. (**1.18**) and:

$$D_1 = \begin{vmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{vmatrix}, \quad D_2 = \begin{vmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{vmatrix}, \quad D_3 = \begin{vmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{vmatrix}$$

Please observe that D_1, D_2, D_3 are obtained by replacing Columns 1, 2, 3, respectively, by the column of the right sides of Eq. (**1.19**).

Cramer's rule in Eq. (**1.19**) can be derived by eliminations similar to those for Eq. (**??**), but it also follows from the general case as explained in Theorem 4c.

> Although *impractical* in computations, they have important engineering applications in **eigenvalue** problems (Chapter 2), differential equations, vector algebra (Chapter **??**), and in other areas.

They can be introduced in numerous equivalent ways. Our definition is particularly for dealing with linear systems.

A determinant of order n is a **scalar** associated with an $n \times n$ matrix[25] $\mathbf{A} = [a_{jk}]$, and is denoted by

$$D = \det \mathbf{A} = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdot & \cdot & \ddots & \vdots \\ \cdot & \cdots & \cdots & \cdot \end{vmatrix} \tag{1.20}$$

[25] This means for there to be a determinant, the said matrix must be of square shape.

For $n = 1$, this determinant is defined by

$$D = a_{11}.$$

For $n \geqslant 2$ the calculation could be either:

$$D = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in} \qquad (j = 1, 2, \ldots, \text{ or } n),$$
$$D = a_{1k}C_{1k} + a_{2k}C_{2k} + \cdots + a_{nk}C_{nk} \qquad (k = 1, 2, \ldots, \text{ or } n). \qquad \text{where} \qquad C_{jk} = (-1)^{j+k}M_{jk}$$

and M_{jk} is a determinant of order $n-1$, namely, the determinant of the submatrix of $\mathbf{A}$ obtained from $\mathbf{A}$ by omitting the row and column of the entry a_{ik}, that is, the jth row and the kth column.

In this way, D is defined in terms of n determinants of order $n-1$, each of which is, in turn, defined in terms of $n-1$ determinants of order $n-2$, and so onuntil we finally arrive at second-order determinants, in which those submatrices consist of single entries whose determinant is defined to be the entry itself.

From the definition it follows that we may expand D by any row or column, that is, choose in Eq. (**??**) or Eq. (**??**), the entries in any row or column, similarly when expanding the C_{ik}'s in Eq. (**??**) or Eq. (**??**), and so on.

This definition is **unambiguous**, meaning, it gives the same value for D no matter which columns or rows we choose in expanding.

Terms used in connection with determinants are taken from matrices. In D we have n^2 **entries** a_{ik}, also n **rows** and n **columns**, and a **main diagonal** on which $a_{11}, a_{22}, \ldots, a_{nn}$ stand. Two terms are new:

M_{jk} is called the **minor** of a_{jk} in D, and C_{jk} the **cofactor** of a_{jk} in D.

For later use we note that, Eq. (**??**) or Eq. (**??**), can also be written in terms of minors

$$D = \sum_{k=1}^{n} (-1)^{j+k} a_{jk} M_{jk} \qquad (j = 1, 2, \cdots, \text{ or } n) \tag{**1.21**}$$

$$D = \sum_{i=1}^{n} (-1)^{j+k} a_{jk} M_{jk} \qquad (k = 1, 2, \ldots, \text{ or } n). \tag{**1.22**}$$

1.23 Example

Minors and Cofactors of a Third-Order Determinant

In Eq. (**1.21**) and Eq. (**1.22**), of the previous section the minors and cofactors of the entries in the first column can be seen directly. For the entries in the second row the minors are

$$M_{21} = \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}, \qquad M_{22} = \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix}, \qquad M_{23} = \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix}$$

and the cofactors are $C_{21} = -M_{21}$, $C_{22} = +M_{22}$, and $C_{23} = -M_{23}$ ■

Expansions of a Third-Order Determinant — 1.24 Example

$$D = \begin{vmatrix} 1 & 3 & 0 \\ 2 & 6 & 4 \\ -1 & 0 & 2 \end{vmatrix} = 1\begin{vmatrix} 6 & 4 \\ 0 & 2 \end{vmatrix} - 3\begin{vmatrix} 2 & 4 \\ -1 & 2 \end{vmatrix} + 0\begin{vmatrix} 2 & 6 \\ -1 & 0 \end{vmatrix}$$

$$= 1(12 - 0) - 3(4 + 4) + 0(0 + 6) = -12.$$

This is the expansion by the first row. The expansion by the third column is

$$D = 0\begin{vmatrix} 2 & 6 \\ -1 & 0 \end{vmatrix} - 4\begin{vmatrix} 1 & 3 \\ -1 & 0 \end{vmatrix} + 2\begin{vmatrix} 1 & 3 \\ 2 & 6 \end{vmatrix} = 0 - 12 + 0 = -12 \quad \blacksquare$$

Determinant of a Triangular Matrix — 1.25 Example

The determinant of a triangular matrix is simply the multiplication of all the diagonal values:

$$\begin{vmatrix} -3 & 0 & 0 \\ 6 & 4 & 0 \\ 1 & 2 & 5 \end{vmatrix} = -3 \cdot 4 \cdot 5 = -60 \quad \blacksquare$$

1.4.3 General Properties of Determinants

There is an attractive way of finding determinants Eq. (1.20) which consists of applying elementary row operations to Eq. (1.20). By doing so we obtain an **upper triangular** determinant (see Sec. 7.1, for definition with "matrix" replaced by "determinant") whose value is then very easy to compute, being just the product of its diagonal entries. This approach is similar (but not the same!) to what we did to matrices in Sec. 7.3.

Interchanging two rows in a determinant introduces a multiplicative factor of -1 to the value of the determinant!

Behavior of an nth-Order Determinant under Elementary Row Operations — 1.12 Theorem

i. Interchange of two rows multiplies the value of the determinant by -1.

ii. Addition of a multiple of a row to another row does not alter the value of the determinant.

iii. Multiplication of a row by a nonzero constant c multiplies the value of the determinant by c.[26]

An interesting property worth mentioning is $\det(c\mathbf{A}) = c^n \det \mathbf{A}$, and **NOT** $c \det \mathbf{A}$).

[26] This holds also when $c = 0$, but no longer gives an elementary row operation.

Further Properties of nth-Order Determinants — 1.13 Theorem

i. - iii in Theory 1.12 hold also for columns. (d) Transposition leaves the value of a determinant unaltered. (e) A zero row or column renders the value of a determinant zero. (f) Proportional rows or columns render the value of a determinant zero. In particular, a determinant with two identical rows or columns has the value zero.

It is quite remarkable that the important concept of the rank of a matrix A, which is the maximum number of linearly independent row or column vectors of A (see Sec. 7.4), can be related to determinants. Here we may assume that rank $A > 0$ because the only matrices with rank 0 are the zero matrices (see Sec. 7.4).

Rank in Terms of Determinants — 1.14 Theorem

Consider an $m \times n$ matrix $\mathbf{A} = [a_{ik}]$:

1. $\mathbf{A}$ has rank $r \geqslant 1$ if and only if $\mathbf{A}$ has an $r \times r$ submatrix with a nonzero determinant.
2. The determinant of any square submatrix with more than r rows, contained in $\mathbf{A}$, if such a matrix exists, has a value equal to zero.

Furthermore, if $m = n$, we have:

3. An $n \times n$ square matrix $\mathbf{A}$ has rank n if and only if $\det \mathbf{A} \neq 0$.

1.4.4 Cramer's Rule

Theory 1.14 opens the way to the classical solution formula for linear systems known as *Cramer's rule*, which gives solutions as quotients of determinants.

Cramer's rule is **NOT** practical in computations for which the methods in Secs, 7.3 and 20.120.3 are suitable. However, Cramer's rule is of theoretical interest in differential equations (Secs. 2.10 and 3.3) and in other theoretical work that has engineering applications.

Cramer's Theorem — 1.15 Theorem

If a linear system of n equations in the same number of unknowns $x_1, \ldots, x_n$:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ \cdots\cdots\cdots\cdots\cdots\cdots &= \cdot \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n \end{aligned} \tag{1.23}$$

has a nonzero coefficient determinant $D = \det A$, the system has precisely one solution. This solution

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is given by the formulas

$$x_1 = \frac{D_1}{D}, \qquad x_2 = \frac{D_2}{D}, \qquad \cdots, \qquad x_n = \frac{D_n}{D}.$$

where D_k is the determinant obtained from D by replacing in D the kth column by the column with the entries $b_1, \ldots, b_n$.

Hence if the system Eq. (**1.23**) is **homogeneous** and $D \neq 0$, it has only the trivial solution $x_1 = 0, x_2 = 0, \ldots, x_n = 0$. If $D = 0$, the homogeneous system also has nontrivial solutions.

Finally, an important application for Cramer's rule dealing with inverse matrices will be given in the next section.

1.5 Inverse of a Matrix using Gauss-Jordan Elimination

In the calculation of inverses we consider **only** square matrices.

The **inverse** of an $n \times n$ matrix $\mathbf{A} = [a_{ik}]$ is denoted by $\mathbf{A}^{-1}$ and is an $n \times n$ matrix such that the following identity holds:

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$$

where $\mathbf{I}$ is the $n \times n$ unit, or the identity, matrix.

- If $\mathbf{A}$ has an inverse, then $\mathbf{A}$ is called a **nonsingular matrix**.
- If $\mathbf{A}$ has no inverse, then $\mathbf{A}$ is called a **singular matrix**.

The Uniqueness of the Inverse — 1.2 Information

If $\mathbf{A}$ has an inverse, the inverse is **unique**.

If both $\mathbf{B}$ and $\mathbf{C}$ are inverses of $\mathbf{A}$, then $AB = I$ and $CA = I$, so that we obtain the uniqueness from

$$\mathbf{B} = \mathbf{IB} = (\mathbf{CA})\,\mathbf{B} = \mathbf{C}\,(\mathbf{AB}) = \mathbf{CI} = \mathbf{C} \quad \blacksquare$$

We will also state that $\mathbf{A}$ has an inverse[27] if and only if it has maximum possible rank n.

[27] Meaning it is non-singular.

Existence of the Inverse — 1.16 Theorem

The inverse $\mathbf{A}^{-1}$ of an $n \times n$ matrix $\mathbf{A}$ exists if and only if rank $A = n$, therefore if and only if det $A \neq 0$. Therefore $\mathbf{A}$ is nonsingular if rank $A = n$, and is singular if rank $A < n$.

1.5.1 Determination of the Inverse by the Gauss-Jordan Method

To actually determine the inverse $\mathbf{A}^{-1}$ of a nonsingular $n \times n$ matrix $\mathbf{A}$, we can use a variant of the Gauss elimination, called the *Gauss-Jordan elimination*.³ The idea of the method is as follows.

Using A, we form n linear systems

$$Ax_{(1)} = e_{(1)}, \ldots, Ax_{(n)} = e_{(n)}$$

where the vectors $\mathbf{e}_{(1)}, \ldots, \mathbf{e}_{(n)}$ are the columns of the $n \times n$ unit matrix $\mathbf{I}$; thus, $\mathbf{e}_{(1)} = \begin{bmatrix} 1 & 0 & \cdots & 0 \end{bmatrix}^T$, $\mathbf{e}_{(2)} =$

$\begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \end{bmatrix}^T$, etc. These are n vector equations in the unknown vectors $\mathbf{x}_{(1)}, \ldots, \mathbf{x}_{(n)}$. We combine them into a single matrix equation

$\mathbf{AX} = \mathbf{I}$, with the unknown matrix $\mathbf{X}$ having the columns $\mathbf{x}_{(1)}, \ldots, \mathbf{x}_{(n)}$. Correspondingly, we combine the n augmented matrices $[\mathbf{A}\ \mathbf{e}_{(1)}], \ldots, [\mathbf{A}\ \mathbf{e}_{(n)}]$ into one wide $n \times 2n$ augmented matrix $\tilde{\mathbf{A}} = [\mathbf{A}\ \mathbf{I}]$. Now multiplication of $\mathbf{AX} = \mathbf{I}$ by $\mathbf{A}^{-1}$ from the left elimination to $\tilde{\mathbf{A}} = [\mathbf{A}\ \mathbf{I}]$. This gives a matrix of the form $[\mathbf{U}\ \mathbf{H}]$ with upper triangular U because the Gauss elimination triangularizes systems. The Gauss-Jordan method reduces U by further elementary row operations to diagonal form, in fact to the unit matrix I. This is done by eliminating the entries of U above the main diagonal and making the diagonal entries all 1 by multiplication (see Example 1). Of course, the method operates on the entire matrix $[\mathbf{U}\ \mathbf{H}]$, transforming $\mathbf{H}$ into some matrix $\mathbf{K}$, hence the entire $[\mathbf{U}\ \mathbf{H}]$ to [I K]. This is the "augmented matrix" of IX = K. Now IX = $X = A^{-1}$, as shown before. By comparison, $K = A^{-1}$, so that we can read A^{-1} directly from $[I\ K]$. The following example illustrates the practical details of the method.

An Example of Calculating the Inverse of a Matrix

Therefore as an example let's calculate the inverse of the matrix:

$$\mathbf{A} = \begin{bmatrix} -1 & 1 & 2 \\ 3 & -1 & 1 \\ -1 & 3 & 4 \end{bmatrix}.$$

We start by applying the *Gauss elimination* to the following $n \times 2n = 3 \times 6$ matrix, where the color ■ always refers to the previous matrix.

$$[\mathbf{A} \quad \mathbf{I}] = \left[\begin{array}{ccc|ccc} -1 & 1 & 2 & 1 & 0 & 0 \\ 3 & -1 & 1 & 0 & 1 & 0 \\ -1 & 3 & 4 & 0 & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{ccc|ccc} -1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 2 & 7 & 3 & 1 & 0 \\ 0 & 2 & 2 & -1 & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{ccc|ccc} -1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 2 & 7 & 3 & 1 & 0 \\ 0 & 0 & -5 & -4 & -1 & 1 \end{array}\right]$$

This is [**UH**] as produced by the Gauss elimination. Now follow the additional Gauss-Jordan steps, reducing U to I, that is, to diagonal form with entries 1 on the main diagonal.

1.5.2 Formulas for Inverses

Since finding the inverse of a matrix is really a problem of solving a system of linear equations, it is clear that Cramer's rule might come into play. And similarly, as Cramer's rule was useful for

theoretical study but not for computation, so too is the explicit formula (4) in the following theorem useful for theoretical considerations but not recommended for actually determining inverse matrices, except for the frequently occurring 2×2 case as given in (4*).

Inverse of a Matrix by Determinants — 1.17 Theorem

Inverse of a Matrix by Determinants The inverse of a nonsingular $n \times n$ matrix $\mathbf{A} = [a_{ik}]$ is given by (4)

$$\mathbf{A}^{-1} = \frac{1}{\det \mathbf{A}}[C_{jk}]^T = \frac{1}{\det \mathbf{A}} \begin{vmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & \ddots & \vdots \end{vmatrix},$$

where C_{jk} is the cofactor of a_{jk} in det A (see Sec. 7.7). (CAUTION! Note well that in A^{-1}, the cofactor C_{jk} occupies the same place as a_{kj} (not a_{jk}) does in A.) In particular, the inverse of

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \quad is \quad \mathbf{A}^{-1} = \frac{1}{\det \mathbf{A}} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}.$$

(4*)

1.5.3 Unusual Properties of Matrix Multiplication. Cancellation Laws

1.5.4 Determinants of Matrix Products

The determinant of a matrix product AB or BA can be written as the product of the determinants of the factors, and it is interesting that $\det AB = \det BA$, although $AB \neq BA$ in general. The corresponding formula (10) is needed occasionally and can be obtained by Gauss-Jordan elimination (see Example 1) and from the theorem just proved.

Determinant of a Product of Matrices — 1.18 Theorem

For any $n \times n$ matrices **A** and **B**,

$$det(AB) = det(BA) = detAdetB.$$

1.6 Vector, and Inner Product Spaces, Linear Transformations

We have captured the essence of vector spaces in Sec. 7.4.

There, we dealt with special vector spaces that arose naturally in the context of matrices and linear systems. The elements of these vector spaces, called **vectors**, satisfied rules (3) and (4) of Sec. 7.1.[28] These special vector spaces were generated by spans, that is, linear combination of finitely many vectors. Furthermore, each such vector had n real numbers as **components**. Review this material before going on.

[28] which were similar to those for numbers.

We can generalize this idea by taking all vectors with n real numbers as components and obtain the very important real n-dimensional vector space $\mathbb{R}^n$. The vectors are known as **real vectors**. Therefore, each vector in $\mathbb{R}^n$ is an ordered n-tuple of real numbers. Now we can consider special values for n. For $n = 2$, we obtain $\mathbb{R}^2$, the vector space of all ordered pairs, which correspond to the vectors in the plane. For $n = 3$, we obtain R^3, the vector space of all ordered triples, which are the vectors in 3-space.[29]

[29] These vectors have wide applications in mechanics, geometry, and calculus and are basic to the engineer and physicist.

Similarly, if we take all ordered n-tuples of **complex numbers** as vectors and complex numbers as scalars, we obtain the **complex vector space** C^n, which we shall consider in Sec. 8.5.

Furthermore, there are other sets of practical interest consisting of matrices, functions, transformations, or others for which addition and scalar multiplication can be defined in an almost natural way so that they too form vector spaces.

It is perhaps not too great an intellectual jump to create, from the **concrete model** R^n, the abstract concept of a real vector space V by taking the basic properties (3) and (4) in Sec. 7.1 as axioms. In this way, the definition of a real vector space arises.

Real Vector Space 1.7 Definition

A nonempty set V of elements $a, b, \cdots$ is called a **real vector space**,[30] and these elements are called vectors[31] if, in V, there are defined two (**2**) algebraic operations[32] as follows:

[30] Sometimes called real linear space.

[31] Regardless of their nature, which will come out from the context or will be left arbitrary.

[32] called vector addition and scalar multiplication.

Vector Addition

associates with every pair of vectors **a** and **b** of V a unique vector of V, called the sum of **a** and **b** and denoted by $\mathbf{a} + \mathbf{b}$, such that the following axioms are satisfied.

Commutatitivity For any two vectors **a** and **b** of V,

$$\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$$

Associativity For any three vectors **a**, **b**, **c** of V,

$$(\mathbf{a} + \mathbf{b}) + \mathbf{c} = \mathbf{a} + (\mathbf{b} + \mathbf{c}) \qquad \text{written } \mathbf{a} + \mathbf{b} + \mathbf{c}.$$

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I.2 .

I.3 There is a unique vector in V, called the zero vector and denoted by 0, such that for every $\mathbf{a}$ in V, $\mathbf{a} + \mathbf{0} = \mathbf{a}$. **I.4** For every $\mathbf{a}$ in V there is a unique vector in V that is denoted by $-\mathbf{a}$ and is such that

$$\mathbf{a} + (-\mathbf{a}) = \mathbf{0}.$$

II. Scalar multiplication. The real numbers are called scalars. Scalar multiplication associates with every $\mathbf{a}$ in V and every scalar c a unique vector of V, called the product of c and a and denoted by ca (or ac) such that the following axioms are satisfied. **II.1** Distributivity. For every scalar c and vectors $\mathbf{a}$ and $\mathbf{b}$ in V, $c(\mathbf{a} + \mathbf{b}) = c\mathbf{a} + c\mathbf{b}$. **II.2** Distributivity. For all scalars c and k and every $\mathbf{a}$ in V, $(c + k)\mathbf{a} = c\mathbf{a} + k\mathbf{a}$. **II.3** Associativity. For all scalars c and k and every $\mathbf{a}$ in V,

$$c(k\mathbf{a}) = (ck)\mathbf{a}$$

II.4 For every $\mathbf{a}$ in V, $1\mathbf{a} = \mathbf{a}$.

If, in the above definition, we take complex numbers as scalars instead of real numbers, we obtain the axiomatic definition of a **complex vector space**.

Take a look at the axioms in the above definition. Each axiom stands on its own:

It is concise, useful, and it expresses a simple property of V.

There are as few axioms as possible and together they express all the desired properties of V![33]

[33] Selecting good axioms is a process of trial and error which often extends over a long period of time. But once agreed upon, axioms become standard such as the ones in the definition of a real vector space.

The following concepts related to a vector space are exactly defined as those given in Sec. 7.4. For example, a **linear combination** of vectors $a_{(1)}, \ldots, a_{(m)}$ in a vector space V is an expression

$$c_1\mathbf{a}_{(1)} + \cdots + c_m\mathbf{a}_m \qquad (c_1, \cdots, c_m \text{ any scalars}).$$

These vectors form a **linearly independent** set if

$$c_1\mathbf{a}_{(1)} + \cdots + c_m\mathbf{a}_{(m)} = \mathbf{0} \tag{1.24}$$

implies that $c_1 = 0, \ldots, c_m = 0$. Otherwise, if Eq. (1.24) also holds with scalars not all zero, the vectors are called **linearly dependent**.

Note that Eq. (1.24) with $m = 1$ is $c\mathbf{a} = \mathbf{0}$ and shows that a single vector $\mathbf{a}$ is linearly independent if and only if $\mathbf{a} \neq \mathbf{0}$.

V has **dimension** n, or is n-dimensional, if it contains a linearly independent set of n vectors, whereas any set of more than n vectors in V is linearly dependent. That set of n linearly independent vectors is called a **basis** for V. Then every vector in V can be written as a linear combination of the basis vectors. Furthermore, for a given basis, this representation is unique.

Vector Space of Matrices 1.26 Example

The real 2×2 matrices form a four-dimensional real vector space. A basis is

$$\mathbf{B}_{11} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad \mathbf{B}_{12} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \mathbf{B}_{21} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{B}_{22} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

because any 2×2 matrix $\mathbf{A} = [a_{jk}]$ has a unique representation:

$$\mathbf{A} = a_{11}\mathbf{B}_{11} + a_{12}\mathbf{B}_{12} + a_{21}\mathbf{B}_{21} + a_{22}\mathbf{B}_{22} \quad \blacksquare$$

Similarly, the real $m \times n$ matrices with fixed m and n form an mn-dimensional vector space.

Vector Space of Polynomials 1.3 Information

The set of all constant, linear, and quadratic polynomials in x together is a vector space of dimension 3 with basis $\{1, x, x^2\}$ under the usual addition and multiplication by real numbers because these two operations give polynomials **NOT** exceeding degree 2.

If a vector space V contains a linearly independent set of n vectors for every n, no matter how large, then V is called **infinite dimensional**, as opposed to a **finite dimensional** (n-dimensional) vector space just defined.

> An example of an infinite dimensional vector space is the space of all continuous functions on some interval $[a, b]$ of the x-axis.

1.6.1 Inner Product Spaces

If $\mathbf{a}$ and $\mathbf{b}$ are vectors in $\mathbb{R}^n$, regarded as column vectors, we can form the product $\mathbf{a}^T\mathbf{b}$.

> This is a 1×1 matrix, which we can identify with its single entry, that is, with a number.

This product is called the **inner product** or **dot product** of $\mathbf{a}$ and $\mathbf{b}$. Other notations for it are $(\mathbf{a}, \mathbf{b})$ and $\mathbf{a} \cdot \mathbf{b}$. Thus

$$\mathbf{a}^T\mathbf{b} = (\mathbf{a}, \mathbf{b}) = \mathbf{a} \cdot \mathbf{b} = \begin{bmatrix} a_1 \cdots a_n \end{bmatrix} \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} = \sum_{i=1}^{n} a_i b_i = a_1 b_1 + \cdots + a_n b_n.$$

We now extend this concept to general real vector spaces by taking basic properties of $(\mathbf{a}, \mathbf{b})$ as axioms for an **abstract inner product** $(\mathbf{a}, \mathbf{b})$ as follows.

Real Inner Product Space 1.8 Definition

If V is finite dimensional, it is actually a so-called Hilbert space; A real vector space V is called a **real inner product space**,or real pre-Hilbert[34] space, if it has the following property. With every pair of vectors **a** and **b** in V there is associated a real number, which is denoted by (a, b) and is called the inner **product** of **a** and **b**, such that the following axioms are satisfied. **I.** For all scalars q_1 and q_2 and all vectors **a**, **b**, **c** in V,

$$(q_1\mathbf{a} + q_2\mathbf{b}, \mathbf{c}) = q_1(\mathbf{a}, \mathbf{c}) + q_2(\mathbf{b}, \mathbf{c})$$

II. For all vectors **a** and **b** in V,

$$(\mathbf{a}, \mathbf{b}) = (\mathbf{b}, \mathbf{a})$$

III. For every **a** in V, $(\mathbf{a}, \mathbf{a}) \geqslant 0$, $(\mathbf{a}, \mathbf{a}) = 0$ if and only if $\mathbf{a} = \mathbf{0}$ (Positive-definiteness).



[34]David Hilbert *(1862 - 1943)* and philosopher of mathematics and one of the most influential mathematicians of all time.

Hilbert discovered and developed a broad range of fundamental ideas including invariant theory, the calculus of variations, commutative algebra, algebraic number theory, the foundations of geometry, spectral theory of operators and its application to integral equations, mathematical physics, and the foundations of mathematics (particularly proof theory). He adopted and defended Georg Cantor's set theory and transfinite numbers. In 1900, he presented a collection of problems that set a course for mathematical research of the 20th century.

Hilbert and his students contributed to establishing rigor and developed important tools used in modern mathematical physics. He was a co-founder of proof theory and mathematical logic.

Vectors whose inner product is zero are called **orthogonal**. The **length** or **norm** of a vector in V is defined by

$$\|\mathbf{a}\| = \sqrt{(\mathbf{a}, \mathbf{a})} \quad (\geqslant 0). \tag{1.25}$$

A vector of norm 1 is called a **unit vector**.

From these axioms and from Eq. (1.25) one can derive the following relations:

$$\big|(\mathbf{a}, \mathbf{b})\big| \leqslant ||\mathbf{a}||||\mathbf{b}|| \qquad \text{(Cauchy-Schwarz Inequality)}$$

$$\|\mathbf{a} + \mathbf{b}\| \leqslant \|\mathbf{a}\| + \|\mathbf{b}\| \qquad \text{(Triangle inequality)}$$

$$\|\mathbf{a} + \mathbf{b}\|^2 + \|\mathbf{a} - \mathbf{b}\|^2 = 2(\|\mathbf{a}\|^2 + \|\mathbf{b}\|^2) \text{(Parallelogram equality)}$$

n-Dimensional Euclidean Space 1.27 Example

R^n with the inner product

$$(\mathbf{a}, \mathbf{b}) = \mathbf{a}^T\mathbf{b} = a_1b_1 + \cdots + a_nb_n$$

where both **a** and **b** are column vectors is called the n-dimensional Euclidean space and is denoted by E^n or again simply by R^n. Axioms IIII hold, as direct calculation shows. Eq. (1.25) gives the **Euclidean norm**:

$$\|\mathbf{a}\| = \sqrt{(\mathbf{a}, \mathbf{a})} = \sqrt{\mathbf{a}^T\mathbf{a}} = \sqrt{a_1^2 + \cdots + a_n^2}$$

An Inner Product for Functions. Function Space 1.28 Example

The set of all real-valued continuous functions $f(x)$, $g(x)$, $\cdots$ on a given interval $\alpha \leqslant x \leqslant \beta$ is a real vector space under the usual addition of functions and multiplication by scalars (real numbers). On this "function space" we can define an inner product by the integral

$$(f, g) = \int_{-\beta}^{\beta} f(x)g(x)dx.$$

Axioms I-III can be verified by direct calculation. Equation (2) gives the norm

$$||f|| = \sqrt{(f, f)} = \sqrt{\int_0^{\beta} f(x)^2 dx}.$$

Our examples give a first impression of the great generality of the abstract concepts of vector spaces and inner product spaces. Further details belong to more advanced courses (on functional analysis, meaning abstract modern analysis; see [GenRef7] listed in App. 1) and cannot be discussed here. Instead we now take up a related topic where matrices play a central role.

1.6.2 Linear Transformations

Let X and Y be any vector spaces. To each vector $\mathbf{x}$ in X we assign a unique vector $\mathbf{y}$ in Y. Then we say that a **mapping**[35] of X into Y is given.

[35] or **transformation** or **operator**

> Such a mapping is denoted by a capital letter, say F.

The vector $\mathbf{y}$ in Y assigned to a vector $\mathbf{x}$ in X is called the **image** of $\mathbf{x}$ under F and is denoted by $F(x)$. F is called a **linear mapping** or **linear transformation** if, for all vectors $\mathbf{v}$ and $\mathbf{x}$ in X and scalars c,

$$\begin{aligned} F(\mathbf{v}+\mathbf{x}) &= F(\mathbf{v}) + F(\mathbf{x}) \\ F(c\mathbf{x}) &= cF(\mathbf{x}). \end{aligned} \tag{1.26}$$

Linear Transformation of Space R^n into Space R^m From now on we let $X = R^n$ and $Y = R^m$. Then any real $m \times n$ matrix $\vec{A} = [a_{ik}]$ gives a transformation of $\mathbb{R}^n$ into $\mathbb{R}^m$,

$$\mathbf{y} = \mathbf{A}\mathbf{x} \tag{1.27}$$

Since $A(\mathbf{u}+\mathbf{x}) = A\mathbf{u} + A\mathbf{x}$ and $A(c\mathbf{x}) = cA\mathbf{x}$, this transformation is **linear**.

We show that, conversely, every linear transformation F of $\mathbb{R}^n$ into $\mathbb{R}^m$ can be given in terms of an $m \times n$ matrix $\mathbf{A}$, after a basis for R^n and a basis for R^m have been chosen.

Let $\mathbf{e}_{(1)}, \ldots, \mathbf{e}_{(n)}$ be any basis for $\mathbb{R}^n$. Then every $\mathbf{x}$ in $\mathbb{R}^n$ has a unique representation

$$\mathbf{x} = x_1\mathbf{e}_{(1)} + \cdots + x_n\mathbf{e}_{(n)}.$$

Since F is linear, this representation implies for the image $F(\mathbf{x})$:

$$F(\mathbf{x}) = F(x_1\mathbf{e}_{(1)} + \cdots + x_n\mathbf{e}_{(n)}) = x_1F(\mathbf{e}_{(1)}) + \cdots + x_nF(\mathbf{e}_{(n)}).$$

Hence F is uniquely determined by the images of the vectors of a basis for $\mathbb{R}^n$. We now choose for R^n the **standard basis**:

$$\mathbf{e}_{(1)} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ \vdots \end{bmatrix}, \quad \mathbf{e}_{(2)} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ \vdots \end{bmatrix}, \quad \cdots, \quad \mathbf{e}_{(n)} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ \vdots \end{bmatrix}$$

where $\mathbf{e}_{(j)}$ has its jth component equal to 1 and all others 0. We show that we can now determine an $m \times n$ matrix $\mathbf{A} = [a_{jk}]$ such that for every $\mathbf{x}$ in $\mathbb{R}^n$ and image $\mathbf{y} = F(\mathbf{x})$ in R^m,

$$\mathbf{y} = F(\mathbf{x}) = \mathbf{A}\mathbf{x}.$$

Indeed, from the image $\mathbf{y}^{(1)} = F(\mathbf{e}_{(1)})$ of $\mathbf{e}_{(1)}$ we get the condition

$$\mathbf{y}^{(1)} = \begin{bmatrix} y_1^{(1)} \\ y_2^{(1)} \\ \vdots \\ \vdots \\ y_n^{(1)} \end{bmatrix} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ a_{21} & \cdots & a_{2n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mm} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

from which we can determine the first column of $\mathbf{A}$, namely $a_{11} = y_1^{(1)}$, $a_{21} = y_2^{(1)}$, $\cdots$, $a_{m1} = y_m^{(1)}$. Similarly, from the image of $\mathbf{e}_{(2)}$ we get the second column of $\mathbf{A}$, and so on ■

We say that **A represents** F, or is a representation of F, with respect to the bases for R^n and R^m. Quite generally, the purpose of a **representation** is the replacement of one object of study by another object whose properties are more readily apparent.

In three-dimensional Euclidean space E^3 the standard basis is usually written $\mathbf{e}_{(1)} = \mathbf{i}$, $\mathbf{e}_{(2)} = \mathbf{j}$, $\mathbf{e}_{(3)} = \mathbf{k}$. Thus,

$$\mathbf{i} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{j} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{k} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

These are the three unit vectors in the positive directions of the axes of the Cartesian **coordinate system in space**, that is, the usual coordinate system with the same scale of measurement on the three mutually perpendicular coordinate axes.

1.29 Example

Linear Transformations

Interpreted as transformations of Cartesian coordinates in the plane, the matrices

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} a & 0 \\ 0 & 1 \end{bmatrix}$$

represent a reflection in the line $x_2 = x_1$, a reflection in the x_1-axis, a reflection in the origin, and a stretch (when $a > 1$, or a contraction when $0 < a < 1$) in the x_1-direction, respectively.

Our discussion preceding Example 5 is simpler than it may look at first sight. To see this, find A representing the linear transformation that maps (x_1, x_2) onto $(2x_1 - 5x_2, 3x_1 + 4x_2)$. **Solution.** Obviously, the transformation is $y_1 = 2x_1 - 5x_2$ $y_2 = 3x_1 + 4x_2$ From this we can directly see that the matrix is

$$\mathbf{A} = \begin{bmatrix} 2 & -5 \\ 3 & 4 \end{bmatrix}. \qquad \text{Check:} \qquad \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 2 & -5 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2x_1 - 5x_2 \\ 3x_1 + 4x_2 \end{bmatrix}.$$

If **a** in Eq. (**1.27**) is square, $n \times n$, then Eq. (**1.27**) maps R^n into R^n. If this **a** is nonsingular, so that A^{-1} exists (see Sec. 7.8), then multiplication of Eq. (**1.27**) by A^{-1} from the left and use of $A^{-1}A = I$ gives the inverse transformation

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{v}. \tag{1.28}$$

It maps every $\mathbf{y} = \mathbf{y}_0$ onto that x, which by Eq. (**1.27**) is mapped onto y_0. The inverse of a linear transformation is itself linear, because it is given by a matrix, as Eq. (**1.28**) shows.

1.6.3 Composition of Linear Transformations

We want to give you a flavor of how linear transformations in general vector spaces work. You will notice, if you read carefully, that definitions and verifications, as shown in Example 1.30, strictly follow the given rules and you can think your way through the material by going in a slow systematic fashion.

The last operation we want to discuss is composition of linear transformations. Let X, Y, W be general vector spaces. As before, let F be a linear transformation from X to Y. Let G be a linear transformation from W to X. Then we denote, by H, the **composition** of F and G, that is,

$$H = F \circ G = FG = F(G),$$

which means we take transformation G and then apply transformation F to it.[36] Now, to give this a more concrete meaning, if we let **w** be a vector in W, then $G(\mathbf{w})$ is a vector in X and $F(G(\mathbf{w}))$ is a vector in Y. Thus, H maps W to Y, and we can write

[36] in that order, meaning, we go from left to right.

$$H(\mathbf{w}) = (F \circ G)(\mathbf{w}) = (FG)(\mathbf{w}) = F(G(\mathbf{w})), \tag{1.29}$$

which completes the definition of composition in a general vector space setting.

But is composition really linear?

To check this we have to verify that H, as defined in Eq. (**1.29**), obeys the two equations of Eq. (**1.26**).

The Composition of Linear Transformations Is Linear — 1.30 Example

To show that H is indeed linear we must show that Eq. (**1.26**) holds. We have, for two (**2**) vectors, namely $\mathbf{w}_1$, $\mathbf{w}_2$ in W:

$$\begin{aligned}
H(\mathbf{w}_1 + \mathbf{w}_2) &= (F \circ G)(\mathbf{w}_1 + \mathbf{w}_2) \\
&= F(G(\mathbf{w}_1 + \mathbf{w}_2)) \\
&= F(G(\mathbf{w}_1) + G(\mathbf{w}_2)) && \text{by linearity of G} \\
&= F(G(\mathbf{w}_1)) + F(G(\mathbf{w}_2)) && \text{by linearity of F} \\
&= (F \circ G)(\mathbf{w}_1) + (F \circ G)(\mathbf{w}_2) && \text{by Eq. (1.29)} \\
&= H(\mathbf{w}_1) + H(\mathbf{w}_2) && \text{by definition of } H
\end{aligned}$$

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Similarly, $H(c\mathbf{w}_2) = (F \circ G)(c\mathbf{w}_2) = F(G(c\mathbf{w}_2)) = F(c(G(\mathbf{w}_2)))$

$$= cF(G(\mathbf{w}_2)) = c(F \circ G)(\mathbf{w}_2) = cH(\mathbf{w}_2).$$

We defined composition as a linear transformation in a general vector space setting and showed that the composition of linear transformations is indeed linear.

Next we want to relate composition of linear transformations to matrix multiplication.

To do so we let $X = \mathbb{R}^n$, $Y = \mathbb{R}^m$, and $W = \mathbb{R}^p$. This choice of particular vector spaces allows us to represent the linear transformations as matrices and form matrix equations, as was done in Eq. (**1.27**). Thus F can be represented by a general real $m \times n$ matrix $\mathbf{A} = [a_{ik}]$ and G by an $n \times p$ matrix $\mathbf{B} = [b_{ik}]$. Then we can write for F, with column vectors $\mathbf{x}$ with n entries, and resulting vector $\mathbf{y}$, with m entries

$$\mathbf{y} = \mathbf{Ax} \tag{1.30}$$

and similarly for G, with column vector $\mathbf{w}$ with p entries,

$$\mathbf{x} = \mathbf{Bw} \tag{1.31}$$

Substituting Eq. (**??**) into Eq. (**1.31**) gives:

$$\mathbf{v} = \mathbf{Ax} = \mathbf{A}(\mathbf{Bw}) = (\mathbf{AB})\mathbf{w} = \mathbf{ABw} = \mathbf{Cw} \qquad \text{where } C = AB.$$

This is Eq. (**1.29**) in a matrix setting, this is, we can define the composition of linear transformations in the Euclidean spaces as multiplication by matrices. Hence, the real $m \times p$ matrix $\mathbf{C}$ represents a linear transformation H which maps R^p to R^n with vector w, a column vector with p entries.

Our discussion is similar to the one in Sec. 7.2, where we motivated the "unnatural" matrix multiplication of matrices. Look back and see that our current, more general, discussion is written out there for the case of dimension $m = 2$, $n = 2$, and $p = 2$. (You may want to write out our development by picking small distinct dimensions, such as $m = 2$, $n = 3$, and $p = 4$, and writing down the matrices and vectors. This is a trick of the trade of mathematicians in that we like to develop and test theories on smaller examples to see that they work.)

1.31 Example

Linear Transformations. Composition

In Example 5 of Sec. 7.9, let **a** be the first matrix and **B** be the fourth matrix with $a > 1$. Then, applying **B** to a vector $\mathbf{w} = [w_1\ w_2]^T$, stretches the element w_1 by a in the x_1 direction. Next, when we apply **A** to the "stretched" vector, we reflect the vector along the line $x_1 = x_2$, resulting in a vector $\mathbf{y} = \begin{bmatrix} w_2 & aw_1 \end{bmatrix}^T$. But this represents, precisely, a geometric description for the composition H of two linear transformations F and G represented by matrices A and B. We now show that, for this example, our result can be obtained by straightforward matrix multiplication, that is, which is the same as before. This shows that indeed $AB = C$, and we see the composition of linear transformations can be represented by a linear transformation. It also shows that the order of matrix multiplication is important (!). You may want to try applying A first and then B, resulting in BA. What do you see? Does it make geometric sense? Is it the same result as AB?

We have learned several abstract concepts such as vector space, inner product space, and linear transformation. The introduction of such concepts allows engineers and scientists to communicate in a concise and common language. For example, the concept of a vector space encapsulated a lot of ideas in a very concise manner. For the student, learning such concepts provides a foundation for more advanced studies in engineering. This concludes Chapter 7. The central theme was the Gaussian elimination of Sec. 7.3 from which most of the other concepts and theory flowed. The next chapter again has a central theme, that is, eigenvalue problems, an area very rich in applications such as in engineering, modern physics, and other areas.

References

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Chapter 2

Linear Algebra II - Eigenvalue Problems

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2.2 Symmetric, Skew-Symmetric, and Orthogonal Matrices 63
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2.4 Complex Matrices and Forms 72

A matrix eigenvalue problem considers the following vector equation:

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x} \tag{2.1}$$

Here $\mathbf{A}$ is a given square matrix, λ an unknown scalar, and $\mathbf{x}$ an unknown vector. In a matrix eigenvalue problem, the task is to determine λ's and $\mathbf{x}$'s that satisfy Eq. (2.15).

> Since $x = 0$ is always a solution for any λ and therefore **NOT** interesting, we only focus on solutions with $x \neq 0$.

The solutions to Eq. (2.15) are given the following names:

> The λ's that satisfy Eq. (2.15) are called **eigenvalues** of $\mathbf{A}$ and the corresponding nonzero $\mathbf{x}$'s that also satisfy Eq. (2.15) are called **eigenvectors** of $\mathbf{A}$.

From this rather innocent looking vector equation flows an amazing amount of relevant theory and an incredible richness of applications.

Applications of Eigenvalues 2.1 Information

Eigenvalue problems come up all the time in engineering, physics, geometry, numerics, theoretical mathematics, biology, environmental science, urban planning, economics, psychology, and other areas.[1]

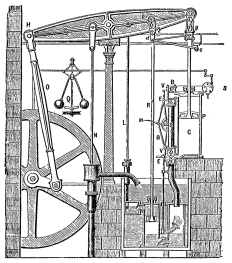
[1] Thus, in your career you are likely to encounter eigenvalue problems.

We start with the fundamentals and describe the concept of eigenvalue problems in Section 2.1 and explain Eq. (**2.15**) with numerous matrices and then we will closely look at different applications where this can be applied ranging from mass-spring systems of physics to population control models of environmental science.

Later on we will investigate eigenvalue problems for real symmetric, skew-symmetric, and orthogonal matrices in in Section 2.2 and their complex counterparts[2] in Section 2.4. Between these section, in Section 2.3, we will have a look at how by diagonalizing a matrix, we obtain its eigenvalues.

[2] which are important in modern physics.

A History of Eigenvalues



It is quite interesting that, eigenvalues are often introduced in the context of *linear algebra* or theory of matrices. However, from a historical standpoint, they were first used in the study of **quadratic forms** and **differential equations**.

In the 18th century, *Leonhard Euler* studied the *rotational motion of a rigid body*, and discovered the importance of the principal axes.[3] *Joseph-Louis Lagrange* realized that the principal axes are the eigenvectors of the inertia matrix [1].

[3] In 1751, he proved that any body has a principal axis of rotation. In his paper "*Du mouvement d'un corps solide quelconque lorsqu'il tourne autour d'un axe mobile*", he states: ··· Whatever be the shape of the body, one can always assign to it such an axis, which passes through its center of gravity, around which it can rotate freely and with a uniform motion.

For example, Movement of each of the components of the Boulton & Watt Steam Engine (1784) can be described as a collection of rigid body elements given the parts do not deform under the action of applied forces.

In the early 19th century, *Augustin-Louis Cauchy* saw how their work could be used to classify the quadric surfaces,[4] and generalized it to arbitrary dimensions [1]. Cauchy also coined the term *racine caractéristique* (characteristic root), for what is now called **eigenvalue** which his term survives in characteristic equation on solving differential equations.

[4] a generalization of conic sections i.e., ellipses, parabolas, and hyperbolas.

Later, *Joseph Fourier* used the work of *Lagrange* and *Pierre-Simon Laplace* to solve the heat equation by separation of variables in his 1822 treatise The Analytic Theory of Heat [2]. *Charles-François Sturm* further elaborated on *Fourier*'s ideas, and brought them to the attention of *Cauchy*, who combined them with his own ideas and arrived at the fact that real symmetric matrices have real eigenvalues [1].

This was extended by Charles Hermite in 1855 to what we call *Hermitian matrices* [2].

Around the same time, *Francesco Brioschi* proved that the eigenvalues of orthogonal matrices lie on the unit circle [1], and *Alfred Clebsch* found the corresponding result for **skew-symmetric matrices** [2].

Finally, Karl Weierstrass clarified an important aspect in the stability theory started by Laplace, by realizing that defective matrices can cause instability [1].

In the meantime, *Joseph Liouville* studied eigenvalue problems similar to those of *Sturm*; the discipline that grew out of their work is now called **SturmLiouville** theory [2].[5] *Karl Hermann Amandus Schwarz* studied the first eigenvalue of *Laplace*'s equation on general domains towards the end of the 19th century, while *Poincaré* studied *Poisson*'s equation a few years later [2].

[5] This topic will be covered in **M.Sc Higher Mathematics II**.

At the start of the 20th century, *David Hilbert* studied the eigenvalues of integral operators by viewing the operators as infinite matrices [2]. He was the first to use the German word **eigen**, which means **own** [3], to denote eigenvalues and eigenvectors[6] though he may have been following a related usage by *Hermann von Helmholtz*. For some time, the standard term in English was "proper value", but the more distinctive term "eigenvalue" became the standard today [4].

[6] In his paper he writes

"Insbesondere in dieser ersten Mitteilung gelange ich zu Formeln, die die Entwickelung einer willkürlichen Funktion nach gewissen ausgezeichneten Funktionen, die ich 'Eigenfunktionen' nenne, liefern: . . ."

(*In particular, in this first report I arrive at formulas that provide the [series] development of an arbitrary function in terms of some distinctive functions, which I call eigenfunctions:* . . .)

"Dieser Erfolg ist wesentlich durch den Umstand bedingt, daß ich nicht, wie es bisher geschah, in erster Linie auf den Beweis für die Existenz der Eigenwerte ausgehe, . . ."

(*This success is fundamentally attributable to the fact that I do not, as it has happened until now, first of all aim at a proof of the existence of eigenvalues* . . .)

Numerical Solving of Eigenvalues The first numerical algorithm for computing eigenvalues and eigenvectors appeared in 1929, when *Richard von Mises* published the **power method** [5]. One of the most popular methods today, the **QR algorithm**, was proposed independently by John G. F. Francis [6] and Vera Kublanovskaya [7] in 1961 [8].

2.1 Determining Eigenvalues and Eigenvectors

We will start simple, and consider multiplying *nonzero vectors* by a given **square matrix**, such as,

$$\begin{bmatrix} 6 & 3 \\ 4 & 7 \end{bmatrix} \begin{bmatrix} 5 \\ 1 \end{bmatrix} = \begin{bmatrix} 33 \\ 27 \end{bmatrix}, \qquad \begin{bmatrix} 6 & 3 \\ 4 & 7 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 30 \\ 40 \end{bmatrix}.$$

We want to see what influence the multiplication of the given matrix has on the vectors.

In the first case, we get a totally new vector with a different direction and different length when compared to the original vector.

> This is what usually happens and is of no interest here.

In the second case something *interesting* happens.

> The multiplication produces a vector $\begin{bmatrix} 30 & 40 \end{bmatrix}^T = 10 \begin{bmatrix} 3 & 4 \end{bmatrix}^T$, which means the new vector has the **same direction as the original vector**.

The **scale constant**, which we denote by λ is 10. The problem of systematically finding such λ's and nonzero vectors for a given square matrix will be the primary focus of this second chapter on Linear Algebra.

> It is called the **matrix eigenvalue problem** or, more commonly, the **eigenvalue problem**.

Now we have the necessary background to formalize our observation. Let $\mathbf{A} = [a_{jk}]$ be a given **nonzero square matrix** of dimension $n \times n$.

Consider the following vector equation:

$$\mathbf{Ax} = \lambda\mathbf{x}$$

The problem of finding nonzero $\mathbf{x}$'s and λ's that satisfy equation Eq. (2.15) is called an **eigenvalue problem**.

In a Nutshell — 2.2 Information

In the aforementioned explanation, $\mathbf{A}$ is a given **square matrix**, $\mathbf{x}$ is an unknown vector, and λ is an unknown scalar. Our task here is to find λ's and nonzero $\mathbf{x}$'s which satisfy Eq. (2.15). Geometrically, we are looking for vectors, $\mathbf{x}$, for which the multiplication by A has the same effect as the multiplication by a scalar λ.[7] Therefore, the multiplication has the effect of producing, from the original vector $\mathbf{x}$, a **new vector** $\lambda\mathbf{x}$ that has the same or opposite, minus sign, direction as the original vector.

You might be wondering why we require $\mathbf{x}$ to be nonzero. Given $\mathbf{x} = \mathbf{0}$ is always a solution of Eq. (2.15) for any value of λ, because $\mathbf{A0} = \mathbf{0}$, it is of no interest and considered **trivial**.

[7] In other words, Ax should be proportional to x.

To complete this introduction to the eigenvalues, let's define some more terminology.

A value of λ, for which Eq. (2.15) has a solution $\mathbf{x} \neq \mathbf{0}$, is called an **eigenvalue** or **characteristic value** of the matrix $\mathbf{A}$.[8] The corresponding solutions $\mathbf{x} \neq 0$ of Eq. (2.15) are called the **eigenvectors** or *characteristic vectors* of $\mathbf{A}$ corresponding to that eigenvalue λ.

[8] Another term for λ is a *latent* root [9]. This term was used to express that eigenvalues are inherently concealed within the matrix's determinant structure.

The set of all the eigenvalues of $\mathbf{A}$ is called the **spectrum** of $\mathbf{A}$ [10].

Usefulness of Spectrum — 2.3 Information

We will see, the **spectrum** consists of at least one eigenvalue and at most of n numerically different eigenvalues. The spectrum provides valuable information about a matrix.

In many applications, such as PageRank [11], one is interested in the dominant eigenvalue.[9]

[9] Which means it is the largest in absolute value which can be found by looking at the spectrum of a matrix.

Finally, the largest of the absolute values of the eigenvalues of $\mathbf{A}$ is called the *spectral radius* of $\mathbf{A}$ [12].

2.1.1 How to Find Eigenvalues and Eigenvectors

Now, with the new terminology for Eq. (2.15) we have in our disposal, we can say that the problem of determining the eigenvalues and eigenvectors of a matrix is called an eigenvalue problem.[10]

[10] However, more precisely, we are considering an *algebraic eigenvalue problem*, as opposed to an eigenvalue problem involving Ordinary Differential Equation (ODE)s or Partial Differential Equation (PDE)s.

As mentioned in the introduction, eigenvalues have a very large number of applications in diverse fields such as in engineering, geometry, physics, mathematics, biology, environmental science, economics, psychology, and other areas.[11]

[11] Throughout this chapter we will encounter applications for elastic membranes, Markov processes, population models, and others.

Given, from the viewpoint of engineering applications, eigenvalue problems are the most important problems in connection with **matrices** and therefore should be closely studied.

We will start with **?? ??** where we demonstrate on how to systematically solve a simple eigenvalue problem.

This example illustrates the general case as follows. Eq. (**2.15**) written in components is:

$$\begin{aligned} a_{11}x_1 + \cdots + a_{1n}x_n &= \lambda x_1 \\ a_{21}x_1 + \cdots + a_{2n}x_n &= \lambda x_2 \\ \cdots\cdots\cdots\cdots\cdots &= \cdots \\ a_{n1}x_1 + \cdots + a_{nn}x_n &= \lambda x_n. \end{aligned}$$

Transferring the terms on the Right Hand Side (RHS) to the Left Hand Side (LHS), we have

$$\begin{aligned} (a_{11} - \lambda)x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= 0 \\ a_{21}x_1 + (a_{22} - \lambda)x_2 + \cdots + a_{2n}x_n &= 0 \\ \cdots\cdots\cdots\cdots\cdots &= \cdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + (a_{nn} - \lambda)x_n &= 0. \end{aligned} \tag{2.2}$$

In matrix notation,

$$(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = 0.$$

By Cramer's theorem in Theory 1.15, this homogeneous linear system of equations has a **nontrivial solution** if and only if the corresponding determinant of the coefficients is zero:

$$D(\lambda) = \det(\mathbf{A} - \lambda\mathbf{I}) = \begin{vmatrix} a_{11} - \lambda & a_{22} & \cdots & a_{1n} \\ a_{21} & a_{22} - \lambda & \ddots & \vdots \\ \vdots & \cdot & \cdots & \cdot \\ a_{n1} & a_{n2} & \cdots & a_{nm} - \lambda \end{vmatrix} = 0 \tag{2.3}$$

Here $\mathbf{A} - \lambda\mathbf{I}$ is called the **characteristic matrix** and $D(\lambda)$ the **characteristic determinant** of $\mathbf{A}$. Eq. (**2.16**) is called the **characteristic equation** of $\mathbf{A}$. By developing $D(\lambda)$ we obtain a polynomial of nth degree in λ.

This is called the **characteristic polynomial** of $\mathbf{A}$.

This proves the following important theorem.

Eigenvalues — 2.1 Theorem

The eigenvalues of a square matrix $\mathbf{A}$ are the **roots** of the characteristic equation Eq. (**2.16**) of $\mathbf{A}$. Therefore an $n \times n$ matrix has **at least one eigenvalue** and at most n numerically different eigenvalues.[12]

[12] For larger n, the actual computation of eigenvalues will, in general, require the use of Newton's method or another numeric approximation method.

The Eigenvalues Needs to be First Determined 2.4 Information

Once these are known, corresponding eigenvectors are obtained from the system Eq. (2.2), for instance, by the **Gauss elimination**, where λ is the eigenvalue for which an eigenvector is wanted. This is what we did in Example 1 and shall do again in the examples below.

Eigenvectors have the following properties.

Eigenvectors and Eigenspace 2.2 Theorem

If $\mathbf{w}$ and $\mathbf{x}$ are eigenvectors of a matrix $\mathbf{A}$ corresponding to the **same** eigenvalue λ, so are $\mathbf{w} + \mathbf{x}$[13] and $k\mathbf{x}$ for any $k \neq 0$.

Therefore, the eigenvectors corresponding to one and the same eigenvalue λ of $\mathbf{A}$, together with $\mathbf{0}$, form a vector space called the **eigenspace** of $\mathbf{A}$ corresponding to that λ.

[13] provided $x \neq -w$.

In particular,

an eigenvector $\mathbf{x}$ is determined only up to a **constant factor**.

Therefore, we can **normalize** $\mathbf{x}$, that is, multiply it by a scalar to get a unit vector.

For example, $\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}^T$ in Example 1 has the length $\|\mathbf{x}_1\| = \sqrt{1^2 + 2^2} = \sqrt{5}$. Therefore $[1/\sqrt{5} \quad 2/\sqrt{5}]^T$ is a normalized eigenvector.[14]

[14] Also known as a unit eigenvector.

Both Example 2.1 and Example 2.2 will illustrate that an $n \times n$ matrix may have n **linearly independent** eigenvectors, or it may have fewer than n. In Example 4 we shall see that a real matrix may have **complex eigenvalues** and eigenvectors.

Multiple Eigenvalues 2.1 Example

Find the eigenvalues and eigenvectors of

$$\mathbf{A} = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}.$$

For our matrix, the characteristic determinant gives the characteristic equation

$$-\lambda^3 - \lambda^2 + 21\lambda + 45 = 0.$$

The roots[15] are $\lambda_1 = 5$, $\lambda_2 = \lambda_3 = -3$.

[15] also known as the eigenvalues of A.

To find eigenvectors, we apply the Gauss elimination to the system $(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}$, first with $\lambda = 5$ and then with $\lambda = -3$.

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For $\lambda = 5$ the characteristic matrix is

$$\mathbf{A} - \lambda\mathbf{I} = \mathbf{A} - 5\mathbf{I} = \begin{bmatrix} -7 & 2 & -3 \\ 2 & -4 & -6 \\ -1 & -2 & -5 \end{bmatrix}. \quad \text{It row-reduces to} \quad \begin{bmatrix} -7 & 2 & -3 \\ 0 & -\frac{24}{7} & -\frac{48}{7} \\ 0 & 0 & 0 \end{bmatrix}.$$

Hence it has rank 2. Choosing $x_3 = -1$ we have $x_2 = 2$ from $-\frac{24}{7}x_2 - \frac{48}{7}x_3 = 0$ and then $x_1 = 1$ from $-7x_1 + 2x_2 - 3x_3 = 0$. Hence an eigenvector of $\mathbf{A}$ corresponding to $\lambda = 5$ is

$$\mathbf{x}_1 = \begin{bmatrix} 1 & 2 & -1 \end{bmatrix}^\mathsf{T}$$

For $\lambda = -3$ the characteristic matrix

$$\mathbf{A} - \lambda\mathbf{I} = \mathbf{A} + 3\mathbf{I} = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -1 & -2 & 3 \end{bmatrix} \quad \text{row-reduces to} \quad \begin{bmatrix} 1 & 2 & -3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Hence it has rank 1. From $x_1 + 2x_2 - 3x_3 = 0$ we have $x_1 = -2x_2 + 3x_3$. Choosing $x_2 = 1$, $x_3 = 0$ and $x_2 = 0, x_3 = 1$, we obtain two linearly independent eigenvectors of A corresponding to $\lambda = -3$,

$$\mathbf{x}_2 = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{x}_3 = \begin{bmatrix} 3 \\ 0 \end{bmatrix} \quad \blacksquare$$

The order M_λ of an eigenvalue λ as a root of the characteristic polynomial is called the **algebraic multiplicity** of λ.[16] The number m_λ of linearly independent eigenvectors corresponding to λ is called the **geometric multiplicity** of λ. Thus m_λ is the dimension of the eigenspace corresponding to this λ.

[16] It basically refers to the number of times a specific eigenvalue is repeated as a root of a matrix's characteristic polynomial. It indicates how "dense" or frequent a particular eigenvalue is when solving for the characteristic equation

Since the characteristic polynomial has degree n, the sum of all the algebraic multiplicities must equal n. In Example 2.1 for $\lambda = -3$ we have $m_\lambda = M_\lambda = 2$. In general,

$$m_\lambda \leqslant M_\lambda,$$

as can be shown. The difference $\Delta_\lambda = M_\lambda - m_\lambda$ is called the **defect** of λ Thus $\Delta_{-3} = 0$ in Example 2.1, but positive defects Δ_λ can easily occur:

2.2 Example

Algebraic Multiplicity, Geometric Multiplicity. Positive Defect

The characteristic equation of the matrix

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad \text{is} \quad \det(\mathbf{A} - \lambda\mathbf{I}) = \begin{bmatrix} -\lambda & 1 \\ 0 & -\lambda \end{bmatrix} = \lambda^2 = 0.$$

Hence $\lambda = 0$ is an eigenvalue of algebraic multiplicity $M_0 = 2$. But its geometric multiplicity is only $m_0 = 1$, since eigenvectors result from $-0x_1 + x_2 = 0$, hence $x_2 = 0$, in the form $[x_1\ 0]^T$. Hence for $\lambda = 0$ the defect is $\Delta_0 = 1$. Similarly, the characteristic equation of the matrix

$$\mathbf{A} = \begin{bmatrix} 3 & 2 \\ 0 & 3 \end{bmatrix} \quad \text{is} \quad \det(\mathbf{A} - \lambda\mathbf{I}) = \begin{bmatrix} 3-\lambda & 2 \\ 0 & 3-\lambda \end{bmatrix} = (3-\lambda)^2 = 0.$$

Hence $\lambda = 3$ is an eigenvalue of algebraic multiplicity $M_3 = 2$, but its geometric multiplicity is only

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$m_3 = 1$, since eigenvectors result from $0x_1 + 2x_2 = 0$ in the form $\begin{bmatrix} x_1 \\ 0 \end{bmatrix}^T$.

Real Matrices with Complex Eigenvalues and Eigenvectors — 2.3 Example

Since real polynomials may have complex roots (which then occur in conjugate pairs), a real matrix may have complex eigenvalues and eigenvectors. For instance, the characteristic equation of the skew-symmetric matrix

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \quad \text{is} \quad \det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} -\lambda & 1 \\ -1 & -\lambda \end{vmatrix} = \lambda^2 + 1 = 0.$$

It gives the eigenvalues $\lambda_1 = i$ $(= \sqrt{-1})$, $\lambda_2 = -i$. Eigenvectors are obtained from $-ix_1 + x_2 = 0$ and $ix_1 + x_2 = 0$, respectively, and we can choose $x_1 = 1$ to get

$$\begin{bmatrix} 1 \\ i \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 \\ -i \end{bmatrix}$$

In the next section we shall need the following simple theorem.

Eigenvalues of the Transpose — 2.3 Theorem

The transpose $\mathbf{A}^T$ of a square matrix $\mathbf{A}$ has the same eigenvalues as $\mathbf{A}$.

2.1.2 Applications of Eigenvalue Problems

Having gained a first impression of matrix eigenvalue problems, we shall illustrate their importance with some typical applications.

Here, we have some typical examples from the wide range of applications of matrix eigenvalue problems. The last example, that is, Example 2.3, shows an application involving vibrating springs and ODEs. It falls into the domain of Chapter **??**, which covers matrix eigenvalue problems related to ODE's modeling mechanical systems and electrical networks.

Example 2.3 is included to keep our discussion independent of Chapter **??**.

Stretching of an Elastic Membrane — 2.4 Example

An elastic membrane in the x_1x_2-plane with boundary circle $x_1^2 + x_2^2 = 1$ (Fig. 160) is stretched so that a point $P : (x_1, x_2)$ goes over into the point $Q : (y_1, y_2)$ given by

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \mathbf{Ax} = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}; \quad \text{in components,} \quad \begin{aligned} y_1 &= 5x_1 + 3x_2 \\ y_2 &= 3x_1 + 5x_2. \end{aligned} \tag{2.4}$$

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Find the **principal directions**, that is, the directions of the position vector **x** of P for which the direction of the position vector **y** of Q is the same or exactly opposite. What shape does the boundary circle take under this deformation?

We are looking for vectors **x** such that $\mathbf{y} = \lambda\mathbf{x}$. Since $\mathbf{y} = \mathbf{A}\mathbf{x}$, this gives $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$, the equation of an eigenvalue problem. In components, $Ax = \lambda x$ is

$$\begin{aligned} 5x_1 + 3x_2 &= \lambda x_1 \\ 3x_1 + (5-\lambda)x_2 &= 0 \end{aligned} \quad \text{or} \quad \begin{aligned} (5-\lambda)x_1 + 3x_2 &= 0 \\ 3x_1 + 5x_2 &= \lambda x_2 \end{aligned}$$

The characteristic equation is

$$\begin{vmatrix} 5-\lambda & 3 \\ 3 & 5-\lambda \end{vmatrix} = (5-\lambda)^2 - 9 = 0.$$

Its solutions are $\lambda_1 = 8$ and $\lambda_2 = 2$. These are the eigenvalues of our problem. For $\lambda = \lambda_1 = 8$, our system (2) becomes $-3x_1 + 3x_2 = 0$, | Solution $x_2 = x_1$, x_1 arbitrary,

$$3x_1 - 3x_2 = 0. for instance, x_1 = x_2 = 1$$

For $\lambda_2 = 2$, our system (2) becomes $3x_1 + 3x_2 = 0$, | Solution $x_2 = -x_1$, x_1 arbitrary,

$$3x_1 + 3x_2 = 0. for instance, x_1 = 1, x_2 = -1$$

We thus obtain as eigenvectors of **A**, for instance, $\begin{bmatrix} 1 \\ 1 \end{bmatrix}^T$ corresponding to λ_1 and $\begin{bmatrix} 1 \\ -1 \end{bmatrix}^T$ corresponding to λ_2 (or a nonzero scalar multiple of these). These vectors make 45ř and 135ř angles with the positive x_1-direction. They give the principal directions, the answer to our problem. The eigenvalues show that in the principal directions the membrane is stretched by factors 8 and 2, respectively; see Fig. 160.

Accordingly, if we choose the principal directions as directions of a new Cartesian u₁u₂-coordinate system, say, with the positive u_1-semi-axis in the first quadrant and the positive u_2-semi-axis in the second quadrant of the x_1x_2-system, and if we set $u_1 = r\cos\phi$, $u_2 = r\sin\phi$, then a boundary point of the unstretched circular membrane has coordinates $\cos\phi$, $\sin\phi$. Hence, after the stretch we have

$$z_1 = 8\cos\phi, \qquad z_2 = 2\sin\phi.$$

Since $\cos^2\phi + \sin^2\phi = 1$, this shows that the deformed boundary is an ellipse (Fig. 160)

$$\frac{z_1^2}{z_2^2} + \frac{z_2^2}{z_2^2} = 1. \tag{2.5}$$

Eigenvalue Problems Arising from Markov Processes — 2.5 Example

Markov processes as considered in Example 13 of Sec. 7.2 lead to eigenvalue problems if we ask for the limit state of the process in which the state vector x is reproduced under the multiplication by

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the stochastic matrix **A** governing the process, that is, $\mathbf{Ax} = \mathbf{x}$. Hence **A** should have the eigenvalue 1, and **x** should be a corresponding eigenvector. This is of practical interest because it shows the long-term tendency of the development modeled by the process.

In that example,

$$\mathbf{A} = \begin{bmatrix} 0.7 & 0.1 & 0 \\ 0.2 & 0.9 & 0.2 \\ 0.1 & 0 & 0.8 \end{bmatrix}. \qquad \text{For the transpose,} \qquad \begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0.1 & 0.9 & 0 \\ 0 & 0.2 & 0.8 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Hence A^T has the eigenvalue 1, and the same is true for A by Theorem 3 in Sec. 8.1. An eigenvector x of A for $\lambda = 1$ is obtained from

$$\mathbf{A} - \mathbf{I} = \begin{bmatrix} -0.3 & 0.1 & 0 \\ 0.2 & -0.1 & 0.2 \\ 0.1 & 0 & -0.2 \end{bmatrix}, \quad \text{row-reduced to} \quad \begin{bmatrix} -\frac{3}{10} & \frac{1}{10} & 0 \\ 0 & -\frac{1}{30} & \frac{1}{5} \\ 0 & 0 & 0 \end{bmatrix}.$$

Taking $x_3 = 1$, we get $x_2 = 6$ from $-x_2/30 + x_3/5 = 0$ and then $x_1 = 2$ from $-3x_1/10 + x_2/10 = 0$. This gives $\mathbf{x} = \begin{bmatrix} 2\&6\&1 \end{bmatrix}^\mathsf{T}$. It means that in the long run, the ratio Commercial:Industrial:Residental will approach 2:6:1, provided that the probabilities given by **A** remain about the same.

2.6 Example

Eigenvalue Problems Arising from Population Models. Leslie Model

The Leslie model describes age-specified population growth, as follows. Let the oldest age attained by the females in some animal population be 9 years. Divide the population into three age classes of 3 years each. Let the Leslie matrix[17] be

$$\mathbf{L} = [l_{jk}] = \begin{bmatrix} 0 & 2.3 & 0.4 \\ 0.6 & 0 & 0 \\ 0 & 0.3 & 0 \end{bmatrix} \tag{2.6}$$

where l_{1k} is the average number of daughters born to a single female during the time she is in age class k, and $l_{j,j-1}$ $(j = 2, 3)$ is the fraction of females in age class $j-1$ that will survive and pass into class j. (a) What is the number of females in each class after 3, 6, 9 years if each class initially consists of 400 females? (b) For what initial distribution will the number of females in each class change by the same proportion? What is this rate of change?

[17] The Leslie matrix is a discrete, age-structured model of population growth named after Patrick H. Leslie and used in population ecology [13, 14]. The Leslie matrix (also called the *Leslie model*) is one of the most well-known ways to describe the growth of populations (and their projected age distribution), in which a population is closed to migration, growing in an unlimited environment, and where only one sex, usually the female, is considered.

(a) Initially, $\mathbf{x}_{(0)}^\mathsf{T} = [400\ 400\ 400]$. After 3 years,

$$\mathbf{x}_{(3)} = \mathbf{L}\mathbf{x}_{(0)} = \begin{bmatrix} 0 & 2.3 & 0.4 \\ 0.6 & 0 & 0 \\ 0 & 0.3 & 0 \end{bmatrix} \begin{bmatrix} 400 \\ 400 \\ 400 \end{bmatrix} = \begin{bmatrix} 1080 \\ 240 \\ 120 \end{bmatrix}$$

Similarly, after 6 years the number of females in each class is given by $\mathbf{x}_{(6)}^\mathsf{T} = (\mathbf{L}\mathbf{x}_{(3)})^\mathsf{T} = [600\ 648\ 72]$, and after 9 years we have $\mathbf{x}_{(9)}^\mathsf{T} = (\mathbf{L}\mathbf{x}_{(6)})^\mathsf{T} = [1519.2 \quad 360 \quad 194.4]$. (b) Proportional change means that we are looking for a distribution vector **x** such that $\mathbf{Lx} = \lambda\mathbf{x}$, where λ the rate of change (growth if $\lambda > 1$, decrease if $\lambda < 1$). The characteristic equation is (develop the

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characteristic determinant by the first column)

$$\det(\mathbf{L}-\lambda\mathbf{I}) = -\lambda^3 - 0.6(-2.3\lambda - 0.3\cdot 0.4) = -\lambda^3 + 1.38\lambda + 0.072 = 0.$$

A positive root is found to be (for instance, by Newton's method, Sec. 19.2) $\lambda = 1.2$. A corresponding eigenvector x can be determined from the characteristic matrix

$$\mathbf{A}-1.2\mathbf{I} = \begin{bmatrix} -1.2 & 2.3 & 0.4 \\ 0.6 & -1.2 & 0 \\ 0 & 0.3 & -1.2 \end{bmatrix}, \quad \text{say,} \quad \mathbf{x} = \begin{bmatrix} 1 \\ 0.5 \\ 0.125 \end{bmatrix}$$

where $x_3 = 0.125$ is chosen, $x_2 = 0.5$ then follows from $0.3x_2 - 1.2x_3 = 0$, and $x_1 = 1$ from $-1.2x_1 + 2.3x_2 + 0.4x_3 = 0$. To get an initial population of 1200 as before, we multiply x by $1200/(1 + 0.5 + 0.125) = 738$.

Answer: Proportional growth of the numbers of females in the three classes will occur if the initial values are 738, 369, 92 in classes 1, 2, 3, respectively. The growth rate will be 1.2 per 3 years ■

2.7 Example

Vibrating System of Two Masses on Two Springs

Mass-spring systems involving several masses and springs can be treated as eigenvalue problems. For instance, the mechanical system in Fig. 161 is governed by the system of ODEs

$$y_1'' = -3y_1 - 2(y_1 - y_2) = -5y_1 + 2y_2$$
$$y_2'' = -2(y_2 - y_1) = 2y_1 - 2y_2$$

where y_1 and y_2 are the displacements of the masses from rest, as shown in the figure, and primes denote derivatives with respect to time t. In vector form, this becomes

$$y'' = \begin{bmatrix} y_1'' \\ y_2'' \end{bmatrix} = \mathbf{Ay} = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \tag{2.7}$$

We try a vector solution of the form

$$\mathbf{y} = \mathbf{x}e^{\omega t} \tag{2.8}$$

This is suggested by a mechanical system of a single mass on a spring (see subsection **??**), whose motion is given by exponential functions, and sines and cosines. Substitution into Eq. (2.18) gives

$$\omega^2\mathbf{x}e^{\omega t} = A\mathbf{x}e^{\omega t}.$$

Dividing by $e^{\omega t}$ and writing $\omega^2 = \lambda$, we see that our mechanical system leads to the eigenvalue problem where

$$\mathbf{Ax} = \lambda\mathbf{x} \qquad \text{where} \qquad \lambda = \omega^2 \tag{2.9}$$

From Example 1 in Section 2.1 we see that $\mathbf{A}$ has the eigenvalues $\lambda_1 = -1$ and $\lambda_2 = -6$. Consequently, $\omega = \pm\sqrt{-1} = \pm i$ and $\sqrt{-6} = \pm i\sqrt{6}$, respectively. Corresponding eigenvectors are

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \qquad \text{and} \qquad \mathbf{x}_2 = \begin{bmatrix} 2 \\ -1 \end{bmatrix} \tag{2.10}$$

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From Eq. (**2.19**) we thus obtain the four complex solutions [see (10), Sec. 2.2]

$$\mathbf{x}_1 e^{\pm it} = \mathbf{x}_1(\cos t \pm i \sin t),$$
$$\mathbf{x}_2 e^{\pm i\sqrt{6}t} = \mathbf{x}_2(\cos \sqrt{6}t \pm i \sin \sqrt{6}t).$$

By addition and subtraction (see Section **??**) we get the four real solutions

$$\mathbf{x}_2 \cos \sqrt{6}t \qquad \mathbf{x}_2 \sin \sqrt{6}t \qquad \mathbf{x}_1 \sin t \qquad \mathbf{x}_1 \cos t$$

A general solution is obtained by taking a linear combination of these,

$$\mathbf{y} = \mathbf{x}_1(a_1 \cos t + b_1 \sin t) + \mathbf{x}_2(a_2 \cos \sqrt{6}t + b_2 \sin \sqrt{6}t)$$

with arbitrary constants a_1, b_1, a_2, b_2.[18] By Eq. (**2.21**), the components of y are

$$y_1 = a_1 \cos t + b_1 \sin t + 2a_2 \cos \sqrt{6}t + 2b_2 \sin \sqrt{6}t$$
$$y_2 = 2a_1 \cos t + 2b_1 \sin t - a_2 \cos \sqrt{6}t - b_2 \sin \sqrt{6}t$$

These functions describe harmonic oscillations of the two masses. Physically, this had to be expected because we have neglected damping.

[18] to which values can be assigned by prescribing initial displacement and initial velocity of each of the two masses

2.2 Symmetric, Skew-Symmetric, and Orthogonal Matrices

We will now consider three (3) classes of real square matrices that, because of their remarkable properties, occur quite frequently in applications.

The first two matrices have already been mentioned in Chapter 1. However, here the goal of current section is to show their remarkable properties.

2.1 Definition

Symmetric, Skew-Symmetric, and Orthogonal Matrices

A **real** square matrix $A = [a_{ik}]$ is called **symmetric** if transposition leaves it unchanged,

$$\mathbf{A}^{\mathsf{T}} = \mathbf{A}, \qquad \text{therefore} \qquad a_{kj} = a_{jk},$$

skew-symmetric if transposition gives the negative of $\mathbf{A}$,

$$\mathbf{A}^{\mathsf{T}} = -\mathbf{A}, \qquad \text{therefore} \qquad a_{kj} = -a_{jk},$$

orthogonal if transposition gives the inverse of $\mathbf{A}$,

$$\mathbf{A}^{\mathsf{T}} = \mathbf{A}^{-1}.$$

2.8 Example

Symmetric, Skew-Symmetric, and Orthogonal Matrices

The matrices

$$\begin{bmatrix} -3 & 1 & 5 \\ 1 & 0 & -2 \\ 5 & -2 & 4 \end{bmatrix}, \quad \begin{bmatrix} 0 & 9 & -12 \\ -9 & 0 & 20 \\ 12 & -20 & 0 \end{bmatrix}, \quad \begin{bmatrix} \frac{2}{3} & \frac{1}{3} & \frac{2}{3} \\ -\frac{2}{3} & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} & -\frac{2}{3} \end{bmatrix}$$

are symmetric, skew-symmetric, and orthogonal, respectively ■

Every **skew-symmetric matrix** has all main diagonal entries **zero**.

Any real square matrix $\mathbf{A}$ may be written as the sum of a symmetric matrix $\mathbf{R}$ and a *skew-symmetric* matrix $\mathbf{S}$, where

$$\mathbf{R} = \frac{1}{2}(\mathbf{A} + \mathbf{A}^{\mathsf{T}}) \qquad \text{and} \qquad \mathbf{S} = \frac{1}{2}(\mathbf{A} - \mathbf{A}^{\mathsf{T}}). \tag{2.11}$$

2.9 Example

Illustration of Eq. (2.16)

$$\mathbf{A} = \begin{bmatrix} 9 & 5 & 2 \\ 2 & 3 & -8 \\ 5 & 4 & 3 \end{bmatrix} = \mathbf{R} + \mathbf{S} = \begin{bmatrix} 9.0 & 3.5 & 3.5 \\ 3.5 & 3.0 & -2.0 \\ 3.5 & -2.0 & 3.0 \end{bmatrix} + \begin{bmatrix} 0 & 1.5 & -1.5 \\ -1.5 & 0 & -6.0 \\ 1.5 & 6.0 & 0 \end{bmatrix} \quad ■$$

Eigenvalues of Symmetric and Skew-Symmetric Matrices — 2.4 Theorem

- The eigenvalues of a symmetric matrix are **real**.
- The eigenvalues of a skew-symmetric matrix are **pure imaginary** or **zero**.

Eigenvalues of Symmetric and Skew-Symmetric Matrices — 2.10 Example

The matrices in Eq. (2.15) and Eq. (2.18) of subsection 2.1.2 are symmetric and have real eigenvalues. The skew-symmetric matrix in Example 2.8 has the eigenvalues $0, -25i$, and $25i$.

The following matrix has the real eigenvalues 1 and 5 but is not symmetric.

$$\begin{bmatrix} 3 & 4 \\ 1 & 3 \end{bmatrix}$$

2.2.1 Orthogonal Transformations and Orthogonal Matrices

Orthogonal transformations are transformations of the form where:

$$\mathbf{y} = \mathbf{A}\mathbf{x} \qquad \text{where } \mathbf{A} \text{ is an orthogonal matrix.}$$

With each vector $\mathbf{x}$ in $\mathbb{R}^n$ such a transformation assigns a vector $\mathbf{y}$ in $\mathbb{R}^n$. For instance, the plane rotation through an angle θ

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

is an *orthogonal transformation*. It can be shown that any orthogonal transformation in the plane or in three-dimensional space is a **rotation**.[19]

[19] possibly combined with a reflection in a straight line or a plane, respectively.

The main reason for the importance of orthogonal matrices is shown in the following theory.

Invariance of Inner Product — 2.5 Theorem

An **orthogonal transformation** preserves the value of the **inner product** of vectors $\mathbf{a}$ and $\mathbf{b}$ in $\mathbb{R}^n$, defined by

$$\mathbf{a} \cdot \mathbf{b} = \mathbf{a}^{\mathsf{T}}\mathbf{b} = [a_1 \quad \cdots \quad a_n] \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}. \tag{2.12}$$

Whereby, for any $\mathbf{a}$ and $\mathbf{b}$ in $\mathbb{R}^n$, orthogonal $n \times n$ matrix $\mathbf{A}$, and $\mathbf{u} = \mathbf{A}\mathbf{a}, \mathbf{v} = \mathbf{A}\mathbf{b}$ we have $\mathbf{u} \cdot \mathbf{v} = \mathbf{a} \cdot \mathbf{b}$.

Hence the transformation also preserves the length or norm of any vector a in R^n given by

$$\|\mathbf{a}\| = \sqrt{\mathbf{a} \cdot \mathbf{a}} = \sqrt{\mathbf{a}^{\mathsf{T}}\mathbf{a}}. \tag{2.13}$$

Orthogonal matrices have additional interesting properties as shown below.

Orthonormality of Column and Row Vectors — 2.6 Theorem

A real square matrix is **orthogonal** if and only if its column vectors $\mathbf{a}_1, \ldots, \mathbf{a}_n$ in addition to its row vectors.[20] orm an orthonormal system, that is,

$$\mathbf{a}_j \cdot \mathbf{a}_k = \mathbf{a}_j^T \mathbf{a}_k = \begin{cases} 0 & \text{if} \quad j \neq k \\ 1 & \text{if} \quad i = k \end{cases} \tag{2.14}$$

[20] f

Determinant of an Orthogonal Matrix — 2.7 Theorem

The **determinant** of an orthogonal matrix has the value $+1$ or -1.

Eigenvalues of an Orthogonal Matrix — 2.8 Theorem

The eigenvalues of an orthogonal matrix **A** are **real** or **complex conjugates** in pairs and have absolute value 1.

Eigenvalues of an Orthogonal Matrix — 2.11 Example

The orthogonal matrix in Example 2.8 has the characteristic equation

$$-\lambda^3 + \frac{2}{3}\lambda^2 + \frac{2}{3}\lambda - 1 = 0.$$

Now one of the eigenvalues must be real, hence $+1$ or -1. Trying, we find -1. Division by $\lambda + 1$ gives $-(\lambda^2 - 5\lambda/3 + 1) = 0$ and the two eigenvalues $(5 + i\sqrt{11})/6$ and $(5 - i\sqrt{11})/6$, which have absolute value 1 ■

2.3 Eigenbases. Diagonalization. Quadratic Forms

Up to now, we have emphasized properties of eigenvalues.

We now turn our attention to the general properties of eigenvectors. Eigenvectors of an $n \times n$ matrix $\mathbf{A}$ may form a basis for R^n. If we are interested in a transformation $\mathbf{y} = \mathbf{A}\mathbf{x}$, such an "eigenbasis" (basis of eigenvectors)-if it exists-is of great advantage because then we can represent any $\mathbf{x}$ in $\mathbb{R}^n$ uniquely as a linear combination of the eigenvectors $\mathbf{x}_1, \ldots, \mathbf{x}_n$, say,

$$\mathbf{x} = c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + \cdots + c_n\mathbf{x}_n.$$

And, denoting the corresponding[21] eigenvalues of the matrix $\mathbf{A}$ by $\lambda_1, \ldots, \lambda_n$, we have $\mathbf{A}\mathbf{x}_j = \lambda_j\mathbf{x}_j$, so that we simply obtain:

$$\begin{aligned}\mathbf{y} =& \mathbf{A}\mathbf{x} = \mathbf{A}(c_1\mathbf{x}_1 + \cdots + c_n\mathbf{x}_n)\\ =& c_1\mathbf{A}\mathbf{x}_1 + \cdots + c_n\mathbf{A}\mathbf{x}_n \qquad (2.15)\\ =& c_1\lambda_1\mathbf{x}_1 + \cdots + c_n\lambda_n\mathbf{x}_n.\end{aligned}$$

This shows that we have decomposed the complicated action of $\mathbf{A}$ on an arbitrary vector $\mathbf{x}$ into a sum of simple actions[22] on the eigenvectors of $\mathbf{A}$.

[21] not necessarily distinct.

[22] i.e., multiplication by scalars

This is the point of an **eigenbasis**.

Now if the n eigenvalues are all different, we do obtain a basis:

Basis of Eigenvectors — 2.9 Theorem

If an $n \times n$ matrix $\mathbf{A}$ has n distinct eigenvalues, then $\mathbf{A}$ has a basis of eigenvectors

$$\mathbf{x}_1, \cdots, \mathbf{x}_n \text{ for } R^n$$

Eigenbasis, Nondistinct Eigenvalues, and Nonexsitence — 2.12 Example

The matrix $\mathbf{A} = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix}$ has a basis of eigenvectors $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ corresponding to the eigenvalues $\lambda_1 = 8$, $\lambda_2 = 2$.[23] Even if not all n eigenvalues are different, a matrix $\mathbf{A}$ may still provide an eigenbasis for R^n. See Example 2.1 in Section 2.1, where $n = 3$. On the other hand, $\mathbf{A}$ may not have enough linearly independent eigenvectors to make up a basis. For instance, $\mathbf{A}$ in Example 2.2 is

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad \text{and has only one eigenvector} \quad \begin{bmatrix} k \\ 0 \end{bmatrix} \quad \text{where} \quad (k \neq 0, \text{ arbitrary}).$$

[23] For more information please look at Example 2.1.

Actually, eigenbases exist under much more general conditions than those in Theory 2.9. An important case is the following.

Symmetric Matrices — 2.10 Theorem

A **symmetric matrix** has an orthonormal basis of eigenvectors for $\mathbb{R}^n$.

Orthonormal Basis of Eigenvectors — 2.13 Example

The first matrix in Example 1 is symmetric, and an orthonormal basis of eigenvectors is $[1/\sqrt{2} \quad 1/\sqrt{2}]^\top$, $[1/\sqrt{2} \; -1/\sqrt{2}]^\top$ ■

2.3.1 Similarity of Matrices and Diagonalization

Eigenbases also play a role in reducing $\mathbf{A}$ to a diagonal matrix whose entries are the eigenvalues of $\mathbf{A}$. This is done by a **similarity transformation**, which is defined as follows:

Similar Matrices, Similarity Transformation — 2.2 Definition

An $n \times n$ matrix $\hat{\mathbf{A}}$ is called **similar** to an $n \times n$ matrix $\mathbf{A}$ if

$$\hat{\mathbf{A}} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P} \tag{2.16}$$

for some, nonsingular, $n \times n$ matrix $\mathbf{P}$.

This transformation, which gives $\hat{\mathbf{A}}$ from $\mathbf{A}$, is called a **similarity transformation**.

The key property of this transformation is that it **preserves the eigenvalues** of $\mathbf{A}$:

Eigenvalues and Eigenvectors of Similar Matrices — 2.11 Theorem

If $\hat{\mathbf{A}}$ is **similar** to $\mathbf{A}$, then $\hat{\mathbf{A}}$ has the **same eigenvalues** as $\mathbf{A}$.

In addition to the previous statement, if $\mathbf{x}$ is an eigenvector of $\mathbf{A}$, then $\mathbf{y} = \mathbf{P}^{-1}\mathbf{x}$ is an eigenvector of $\hat{\mathbf{A}}$ corresponding to the same eigenvalue.

Eigenvalues and Vectors of Similar Matrices — 2.14 Example

Let

$$\mathbf{A} = \begin{bmatrix} 6 & -3 \\ 4 & -1 \end{bmatrix} \quad \text{and} \quad \mathbf{P} = \begin{bmatrix} 1 & 3 \\ 1 & 4 \end{bmatrix}.$$

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Then

$$\hat{\mathbf{A}} = \begin{bmatrix} 4 & -3 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 6 & -3 \\ 4 & -1 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}.$$

Here $\mathbf{P}^{-1}$ was obtained from (4*) in Sec. 7.8 with $\det \mathbf{P} = 1$. We see that $\hat{\mathbf{A}}$ has the eigenvalues $\lambda_1 = 3$, $\lambda_2 = 2$. The characteristic equation of $\mathbf{A}$ is $(6 - \lambda)(-1 - \lambda) + 12 = \lambda^2 - 5\lambda + 6 = 0$. It has the roots (the eigenvalues of $\mathbf{A}$) $\lambda_1 = 3$, $\lambda_2 = 2$, confirming the first part of Theory 2.11.

We confirm the second part. From the first component of $(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}$ we have $(6\lambda)x_1 3x_2 = 0$. For $\lambda = 3$ this gives $3x_1 - 3x_2 = 0$, say $\mathbf{x}_1 = [1 \quad 1]^T$. For $\lambda = 2$, it gives, $4x_1 - 3x_2 = 0$, say, $\mathbf{x}_2 = [3 \quad 4]^T$. In Theorem 3 we therefore have;

$$\mathbf{y}_1 = \mathbf{P}^{-1}\mathbf{x}_1 = \begin{bmatrix} 4 & -3 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad \mathbf{y}_2 = \mathbf{P}^{-1}\mathbf{x}_2 = \begin{bmatrix} 4 & -3 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Indeed, these are eigenvectors of the diagonal matrix $\hat{\mathbf{A}}$.

Perhaps we see that $\mathbf{x}_1$ and $\mathbf{x}_2$ are the columns of $\mathbf{P}$. This suggests the general method of transforming a matrix $\hat{\mathbf{A}}$ to diagonal form $\mathbf{D}$ by using $\mathbf{P} = \mathbf{X}$, the matrix with eigenvectors as columns.

By a suitable similarity transformation we can now transform a matrix $\mathbf{A}$ to a diagonal matrix $\mathbf{D}$ whose diagonal entries are the eigenvalues of $\mathbf{A}$:

Diagonalization of a Matrix — 2.12 Theorem

If an $n \times n$ matrix $\mathbf{A}$ has a **basis of eigenvectors**, then

$$\mathbf{D} = \mathbf{X}^{-1}\mathbf{A}\mathbf{X} \tag{2.17}$$

is **diagonal**, with the eigenvalues of $\mathbf{A}$ as the entries on the main diagonal. Here $\mathbf{X}$ is the matrix with these eigenvectors as column vectors. Also,

$$\mathbf{D}^m = \mathbf{X}^{-1}\mathbf{A}^m\mathbf{X} \qquad \text{where} \qquad m = 2, 3, \cdots$$

Application of Diagonalization — 2.15 Example

Diagonalize

$$\mathbf{A} = \begin{bmatrix} 7.3 & 0.2 & -3.7 \\ -11.5 & 1.0 & 5.5 \\ 1.7 & 1.2 & 0.2 \end{bmatrix}.$$

The characteristic determinant gives the characteristic equation $-\lambda^3 - \lambda^2 + 12\lambda = 0$. The roots (eigenvalues of $\mathbf{A}$) are $\lambda_1 = 3$, $\lambda_2 = -4$, $\lambda_3 = 0$. By the Gauss elimination applied to $(\mathbf{A} - \lambda I)x = 0$ with $\lambda = \lambda_1, \lambda_2, \lambda_3$ we find eigenvectors and then $\mathbf{X}^{-1}$ by the GaussJordan elimination (Sec. 7.8,

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Example 1). The results are:

$$\begin{bmatrix}1\\3\\1\end{bmatrix}, \begin{bmatrix}1\\-1\\3\end{bmatrix}, \begin{bmatrix}2\\1\\4\end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix}-1 & 1 & 2\\3 & -1 & 1\\-1 & 3 & 4\end{bmatrix}, \quad \mathbf{X}^{-1} = \begin{bmatrix}-0.7 & 0.2 & 0.3\\-1.3 & -0.2 & 0.7\\0.8 & 0.2 & -0.2\end{bmatrix}.$$

Calculating $\mathbf{AX}$ and multiplying by $\mathbf{X}^{-1}$ from the left, we thus obtain

$$\mathbf{D} = \mathbf{X}^{-1}\mathbf{AX} = \begin{bmatrix}-0.7 & 0.2 & 0.3\\-1.3 & -0.2 & 0.7\\0.8 & 0.2 & -0.2\end{bmatrix}\begin{bmatrix}-3 & -4 & 0\\9 & 4 & 0\\-3 & -12 & 0\end{bmatrix} = \begin{bmatrix}3 & 0 & 0\\0 & -4 & 0\\0 & 0 & 0\end{bmatrix} \quad \blacksquare$$

2.3.2 Quadratic Forms and Transformation to Principal Axes

By definition, a **quadratic form** Q in the components $x_1, \dots, x_n$ of a vector $\mathbf{x}$ is a sum of n^2 terms, namely,

$$\begin{aligned} Q =& \mathbf{x}^\mathsf{T}\mathbf{A}\mathbf{x} = \sum_{i=1}^{n}\sum_{k=1}^{n} a_{jk}x_j x_k \\ =& a_{11}x_1^2 + a_{12}x_1x_2 + \cdots + a_{1n}x_1x_n \\ &+ a_{21}x_2x_1 + a_{22}x_2^2 + \cdots + a_{2n}x_2x_n \\ &\cdots \\ &a_{n1}x_nx_1 + a_{n2}x_nx_2 + \cdots + a_{nn}x_n^2 \end{aligned} \tag{2.18}$$

$A = [a_{jk}]$ is called the **coefficient matrix** of the form. We may assume that A is symmetric, because we can take off-diagonal terms together in pairs and write the result as a sum of two equal terms; see the following example.

Quadratic Form and Symmetric Coefficient Matrix — 2.16 Example

$$\mathbf{x}^\mathsf{T}\mathbf{A}\mathbf{x} = \begin{bmatrix}x_1 & x_2\end{bmatrix}\begin{bmatrix}3 & 4\\6 & 2\end{bmatrix}\begin{bmatrix}x_1\\x_2\end{bmatrix} = 3x_1^2 + 4x_1x_2 + 6x_2x_1 + 2x_2^2 = 3x_1^2 + 10x_1x_2 + 2x_2^2.$$

Here $4+6 = 10 = 5+5$. From the corresponding **symmetric** matrix $C = [c_{ik}]$, where $c_{ik} = \frac{1}{2}(a_{ik}+a_{ki})$, thus $c_{11} = 3$, $c_{12} = c_{21} = 5$, $c_{22} = 2$, we get the same result; indeed,

$$\mathbf{x}^\mathsf{T}\mathbf{C}\mathbf{x} = \begin{bmatrix}x_1 & x_2\end{bmatrix}\begin{bmatrix}3 & 5\\5 & 2\end{bmatrix}\begin{bmatrix}x_1\\x_2\end{bmatrix} = 3x_1^2 + 5x_1x_2 + 5x_2x_1 + 2x_2^2 = 3x_1^2 + 10x_1x_2 + 2x_2^2.$$

Quadratic forms occur in physics and geometry, for instance, in connection with conic sections (ellipses $x_1^2/a^2 + x_2^2/b^2 = 1$, etc.) and quadratic surfaces (cones, etc.). Their transformation to principal axes is an important practical task related to the diagonalization of matrices, as follows.

By Theory 2.10, the symmetric coefficient matrix $\mathbf{A}$ of Eq. (2.18) has an orthonormal basis of eigenvectors. Hence if we take these as column vectors, we obtain a matrix $\mathbf{X}$ that is orthogonal, so that $\mathbf{X}^{-1} = \mathbf{X}^\mathsf{T}$. From Eq. (2.17) we thus have $\mathbf{A} = \mathbf{X}\mathbf{D}\mathbf{X}^{-1} = \mathbf{X}\mathbf{D}\mathbf{X}^\mathsf{T}$. Substitution into Eq.

(2.18) gives

$$Q = \mathbf{x}^\mathsf{T}\mathbf{X}\mathbf{D}\mathbf{X}^\mathsf{T}\mathbf{x}. \tag{2.19}$$

If we set $\mathbf{X}^\mathsf{T}\mathbf{x} = \mathbf{y}$, then, since $\mathbf{X}^{\mathsf{T}x} = \mathbf{X}^{-1}$, we have $\mathbf{X}^{-1}\mathbf{x} = \mathbf{y}$ and thus obtain

$$\mathbf{x} = \mathbf{X}\mathbf{y} \tag{2.20}$$

Furthermore, in Eq. (2.19) we have $\mathbf{x}^\mathsf{T}\mathbf{X} = (\mathbf{X}^\mathsf{T}\mathbf{x})^\mathsf{T} = \mathbf{y}^\mathsf{T}$ and $\mathbf{X}^\mathsf{T}\mathbf{x} = \mathbf{y}$, so that Q becomes simply

$$Q = \mathbf{y}^\mathsf{T}\mathbf{D}\mathbf{y} = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \cdots + \lambda_n y_n^2. \tag{2.21}$$

This proves the following basic theorem.

Principal Axes Theorem — 2.13 Theorem

The substitution Eq. (2.20) transforms a quadratic form

$$Q = \mathbf{x}^\mathsf{T}\mathbf{A}\mathbf{x} = \sum_{i=1}^{n}\sum_{k=1}^{n} a_{jk}x_j x_k \qquad \text{where} \qquad (a_{kj} = a_{jk})$$

to the principal axes form or canonical form Eq. (2.21), where $\lambda_1, \ldots, \lambda_n$ are the eigenvalues[24] of the symmetric matrix $\mathbf{A}$, and $\mathbf{X}$ is an orthogonal matrix with corresponding eigenvectors $\mathbf{x}_1, \ldots, \mathbf{x}_n$, respectively, as column vectors.

[24] Which are not necessarily distinct.

Transformation to Principal Axes and Conic Sections — 2.17 Example

Find out what type of conic section the following quadratic form represents and transform it to principal axes:

$$Q = 17x_1^2 - 30x_1x_2 + 17x_2^2 = 128.$$

We have $Q = \mathbf{x}^\mathsf{T}\mathbf{A}\mathbf{x}$, where

$$\mathbf{A} = \begin{bmatrix} 17 & -15 \\ -15 & 17 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

This gives the characteristic equation $(17-\lambda)^2 - 15^2 = 0$. It has the roots $\lambda_1 = 2$, $\lambda_2 = 32$. Hence Eq. (2.21) becomes

$$Q = 2y_1^2 + 32y_2^2.$$

We see that $Q = 128$ represents the ellipse $2y_1^2 + 32y_2^2 = 128$, that is,

$$\frac{y_1^2}{y_2^2} + \frac{y_2^2}{y_2^2} = 1.$$

If we want to know the direction of the principal axes in the x_1x_2-coordinates, we have to determine normalized eigenvectors from $(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = 0$ with $\lambda = \lambda_1 = 2$ and $\lambda = \lambda_2 = 32$ and then use Eq. (2.20). We get

$$\begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix},$$

Continues on next page →

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Therefore

$$\mathbf{x} = \mathbf{X}\mathbf{y} = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}, \qquad \begin{aligned} x_1 &= y_1/\sqrt{2} - y_2/\sqrt{2} \\ x_2 &= y_1/\sqrt{2} + y_2/\sqrt{2}. \end{aligned}$$

This is a 45° rotation ■

2.4 Complex Matrices and Forms

The three classes of matrices in Section 2.2 have complex counterparts which are of practical interest in certain applications, for instance, in **quantum mechanics**. This is mainly because of their spectra as shown in Theorem 1 in this section. The second topic is about extending quadratic forms of Section 2.3 to complex numbers.

Notations — 2.3 Definition

$\overline{\mathbf{A}} = [\overline{a}_{jk}]$ is obtained from $\mathbf{A} = [a_{jk}]$ by replacing each entry $a_{jk} = \alpha + i\beta$ (α, β real) with its complex conjugate $\overline{a}_{jk} = \alpha - i\beta$. Also, $\overline{\mathbf{A}}^{\mathsf{T}} = [\overline{a}_{kj}]$ is the transpose of $\mathbf{A}$, hence the conjugate transpose of $\mathbf{A}$.

Notations — 2.18 Example

If $\mathbf{A} = \begin{bmatrix} 3+4i & 1-i \\ 6 & 2-5i \end{bmatrix}$, then $\overline{\mathbf{A}} = \begin{bmatrix} 3-4i & 1+i \\ 6 & 2+5i \end{bmatrix}$ and $\overline{\mathbf{A}}^{\mathsf{T}} = \begin{bmatrix} 3-4i & 6 \\ 1+i & 2+5i \end{bmatrix}$ ■

Hermitian, Skew-Hermitian, and Unitary Matrices — 2.4 Definition

A square matrix $\mathbf{A} = [a_{ki}]$ is called

Hermitian	if $\overline{\mathbf{A}}^{\mathsf{T}} = \mathbf{A}$	that is, $\overline{a}_{kj} = a_{jk}$
skew-Hermitian	if $\overline{\mathbf{A}}^{\mathsf{T}} = -\mathbf{A}$,	that is, $\overline{a}_{kj} = -a_{jk}$
unitary	if $\overline{\mathbf{A}}^{\mathsf{T}} = \mathbf{A}^{-1}$.	

The first two classes are named after Charles Hermite.[25]



[25] Charles Hermite *(1822 - 1901)* was a French mathematician who studied analysis, number theory, and algebra. One of his most remarkable achievements was the proof of the transcendence of the number e.

From the definitions we see the following properties. If $\mathbf{A}$ is *Hermitian*, the entries on the main diagonal must satisfy $\overline{a}_{jj} = a_{jj}$; that is, they are **real**. Similarly, if $\mathbf{A}$ is *skew-Hermitian*, then $\overline{a}_{jj} = -a_{jj}$.

If we set $a_{jj} = \alpha + i\beta$, this becomes $\alpha - i\beta = -(\alpha + i\beta)$. Hence $\alpha = 0$, so that a_{jj} must be pure imaginary or 0.

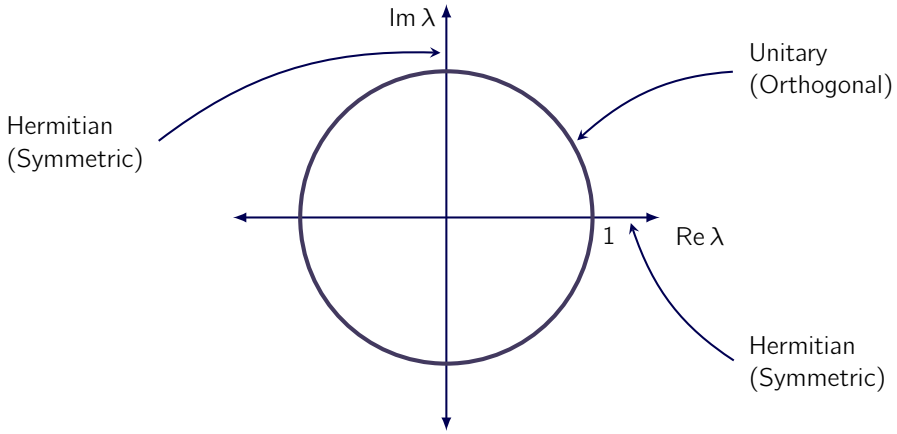
Hermitian, Skew-Hermitian, and Unitary Matrices — 2.19 Example

$$\mathbf{A} = \begin{bmatrix} 4 & 1-3i \\ 1+3i & 7 \end{bmatrix} \qquad \mathbf{B} = \begin{bmatrix} 3i & 2+i \\ -2+i & -i \end{bmatrix} \qquad \mathbf{C} = \begin{bmatrix} \frac{1}{2}i & \frac{1}{2}\sqrt{3} \\ \frac{1}{2}\sqrt{3} & \frac{1}{2}i \end{bmatrix}$$

are Hermitian, skew-Hermitian, and unitary matrices, respectively, as you may verify by using the definitions ■

If a Hermitian matrix is real, then $\overline{\mathbf{A}}^{\mathsf{T}} = \mathbf{A}^{\mathsf{T}} = \mathbf{A}$. Hence a real Hermitian matrix is a symmetric

Figure 2.1
Location of the eigenvalues of Hermitian, skew-Hermitian, and unitary matrices in the complex λ-plane.



matrix (see Section 2.2). Similarly, if a skew-Hermitian matrix is real, then $\overline{\mathbf{A}}^{\mathsf{T}} = \mathbf{A}^{\mathsf{T}} = -\mathbf{A}$. Hence a real skew-Hermitian matrix is a skew-symmetric matrix. Finally, if a unitary matrix is real, then $\overline{\mathbf{A}}^{\mathsf{T}} = \mathbf{A}^{\mathsf{T}} = \mathbf{A}^{\mathsf{T}}$. Hence a real unitary matrix is an orthogonal matrix.

This shows that Hermitian, skew-Hermitian, and unitary matrices generalize symmetric, skew-symmetric, and orthogonal matrices, respectively.

2.4.1 Eigenvalues

It is quite remarkable that the matrices under consideration have spectra[26] which can be characterized in a general way as follows:

[26] This means it has a set of eigenvalues.

Eigenvalues — 2.14 Theorem

- The eigenvalues of a Hermitian matrix, and thus of a symmetric matrix, are real.
- The eigenvalues of a skew-Hermitian matrix, and thus of a skew-symmetric matrix, are pure imaginary or zero.
- The eigenvalues of a unitary matrix, and thus of an orthogonal matrix, have absolute value 1.

Key properties of orthogonal matrices[27] generalize to unitary matrices in a remarkable way.

[27] which are invariance of the inner product, and the orthonormality of rows and columns.

To see this, instead of R^n we now use the **complex vector space** C^n of all complex vectors with n complex numbers as components, and complex numbers as scalars. For such complex vectors the **inner product** is defined by:

$$\mathbf{a} \cdot \mathbf{b} = \overline{\mathbf{a}}^{\mathsf{T}} \mathbf{b}, \tag{2.22}$$

where the **length** or **norm** of such a complex vector is a *real* number defined by

$$\|\mathbf{a}\| = \sqrt{\mathbf{a}\cdot\mathbf{a}} = \sqrt{\bar{\mathbf{a}}_j^{\mathsf{T}}\mathbf{a}} = \sqrt{\bar{a}_1 a_1 + \cdots + \bar{a}_n a_n} = \sqrt{|a_1|^2 + \cdots + |a_n|^2}. \tag{2.23}$$

2.15 Theorem

Invariance of the Inner Product

A unitary transformation, that is, $\mathbf{y} = \mathbf{A}\mathbf{x}$ with a unitary matrix $\mathbf{A}$, preserves the value of the inner product Eq. (2.22), and therefore also the norm Eq. (2.23).

The complex analog of an orthonormal system of real vectors (see Section 2.2) is defined as follows.

2.5 Definition

Unitary System

A unitary system is a set of complex vectors satisfying the relationships

$$\mathbf{a}_j \cdot \mathbf{a}_k = \bar{\mathbf{a}}_j^{\mathsf{T}}\mathbf{a}_k = \begin{cases} 0 & \text{if} \quad j \neq k \\ 1 & \text{if} \quad j = k. \end{cases}$$

Theory 2.6 in Section 2.2 extends to complex as follows.

2.16 Theorem

Unitary Systems of Column and Row Vectors

A complex square matrix is **unitary** if and only if its column vectors[28] form a unitary system.

[28] and also its row vectors.

2.17 Theorem

Determinant of a Unitary Matrix

Let $\mathbf{A}$ be a unitary matrix. Then its determinant has absolute value one, that is,

$$|\det \mathbf{A}| = 1.$$

2.20 Example

A Study on Unitary Matrix

For the vectors $\mathbf{a}^{\mathsf{T}} = \begin{bmatrix} 2 & -i \end{bmatrix}$ and $\mathbf{b}^{\mathsf{T}} = \begin{bmatrix} 1+i & 4i \end{bmatrix}$ we get $\bar{\mathbf{a}}^{\mathsf{T}} = \begin{bmatrix} 2 & i \end{bmatrix}^{\mathsf{T}}$ and $\bar{\mathbf{a}}^{\mathsf{T}}\mathbf{b} = 2(1+i) - 4 = -2 + 2i$ and with

$$\mathbf{A} = \begin{bmatrix} 0.8i & 0.6 \\ 0.6 & 0.8i \end{bmatrix} \quad \text{also} \quad \mathbf{A}\mathbf{a} = \begin{bmatrix} i \\ 2 \end{bmatrix} \quad \text{and} \quad \mathbf{A}\mathbf{b} = \begin{bmatrix} -0.8 + 3.2i \\ -2.6 + 0.6i \end{bmatrix},$$

as one can readily verify. This gives $(\overline{\mathbf{A}}\overline{\mathbf{a}})^{\mathsf{T}}\mathbf{A}\mathbf{b} = -2 + 2i$, illustrating Theory 2.15. The matrix is **unitary**. Its columns form a unitary system,

$$\bar{\mathbf{a}}_1^{\mathsf{T}}\mathbf{a}_1 = -0.8i \cdot 0.8i + 0.6^2 = 1, \qquad \bar{\mathbf{a}}_1^{\mathsf{T}}\mathbf{a}_2 = -0.8i \cdot 0.6 + 0.6 \cdot 0.8i = 0,$$

$$\bar{\mathbf{a}}_2^{\mathsf{T}}\mathbf{a}_2 = 0.6^2 + (-0.8i)0.8i = 1$$

and so do its rows. Also, $\det \mathbf{A} = -1$. The eigenvalues are $0.6 + 0.8i$ and $-0.6 + 0.8i$, with eigenvectors $\begin{bmatrix} 1 \\ 1 \end{bmatrix}^{\mathsf{T}}$ and $\begin{bmatrix} 1 & -1 \end{bmatrix}^{T}$, respectively.

Theory 2.10 in Section 2.3 on the existence of an eigen-basis extends to complex matrices as follows.

Basis of Eigenvectors — 2.18 Theorem

A **Hermitian**, **skew-Hermitian**, or **unitary matrix** has a basis of eigenvectors for C^n that is a **unitary system**.

2.4.2 Hermitian and Skew-Hermitian Forms

The concept of a quadratic form Section 2.3 can be extended to complex. We call the numerator $\bar{\mathbf{x}}^\mathsf{T}\mathbf{A}\mathbf{x}$ in Eq. (2.24) a **form** in the components $x_1, \ldots, x_n$ of $\mathbf{x}$, which may now be complex. This form is again a sum of n^2 terms

$$\begin{aligned}\bar{\mathbf{x}}^\mathsf{T}\mathbf{A}\mathbf{x} &= \sum_{i=1}^{n}\sum_{k=1}^{n} a_{jk}\bar{x}_j x_k \\ &= a_{11}\bar{x}_1 x_1 + \cdots + a_{1n}\bar{x}_1 x_n \\ &\quad + a_{21}\bar{x}_2 x_1 + \cdots + a_{2n}\bar{x}_2 x_n \\ &\quad + \cdots \\ &\quad + a_{n1}\bar{x}_n x_1 + \cdots + a_{nn}\bar{x}_n x_n\end{aligned} \tag{2.24}$$

$\mathbf{A}$ is called its **coefficient matrix**. The form is called a **Hermitian** or **skew-Hermitian** form if $\mathbf{A}$ is Hermitian or skew-Hermitian, respectively. The value of a Hermitian form is **real**, and that of a skew-Hermitian form is **pure imaginary or zero**.

Hermitian Form — 2.21 Example

For $\mathbf{A}$ in Example 2.19 and, say, $\mathbf{x} = \begin{bmatrix} 1+i & 5i \end{bmatrix}^\mathsf{T}$ we get

$$\begin{aligned}\bar{\mathbf{x}}^\mathsf{T}\mathbf{A}\mathbf{x} &= \begin{bmatrix} 1-i & -5i \end{bmatrix}\begin{bmatrix} 4 & 1-3i \\ 1+3i & 7 \end{bmatrix}\begin{bmatrix} 1+i \\ 5i \end{bmatrix} \\ &= \begin{bmatrix} 1-i & -5i \end{bmatrix}\begin{bmatrix} 4(1+i)+(1-3i)\cdot 5i \\ (1+3i)(1+i)+7\cdot 5i \end{bmatrix} = 223 \quad \blacksquare\end{aligned}$$

Clearly, if $\mathbf{A}$ and $\mathbf{x}$ in Eq. (2.22) are real, then Eq. (2.24) reduces to a quadratic form, as discussed in the previous section.

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Acronyms

L

LHS Left Hand Side
Left Hand Side

O

ODE Ordinary Differential Equation
Ordinary Differential Equation

P

PDE Partial Differential Equation
Partial Differential Equation

R

RHS Right Hand Side
Right Hand Side



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